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Fan et al.

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(54) **SEMICONDUCTOR STRUCTURE AND METHOD FOR FORMING THE SAME**

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CPC **H10D 30/0243** (2025.01); **H10D 30/6211** (2025.01); **H10D 30/6219** (2025.01); **H10D 62/151** (2025.01); **H10D 64/01** (2025.01); **H10D 64/62** (2025.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

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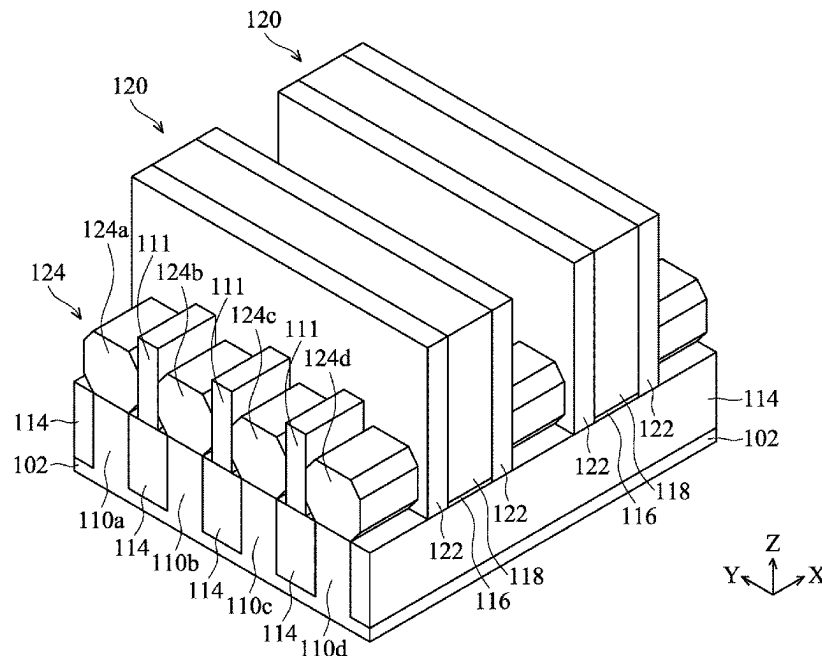
Primary Examiner — William C Trapanese

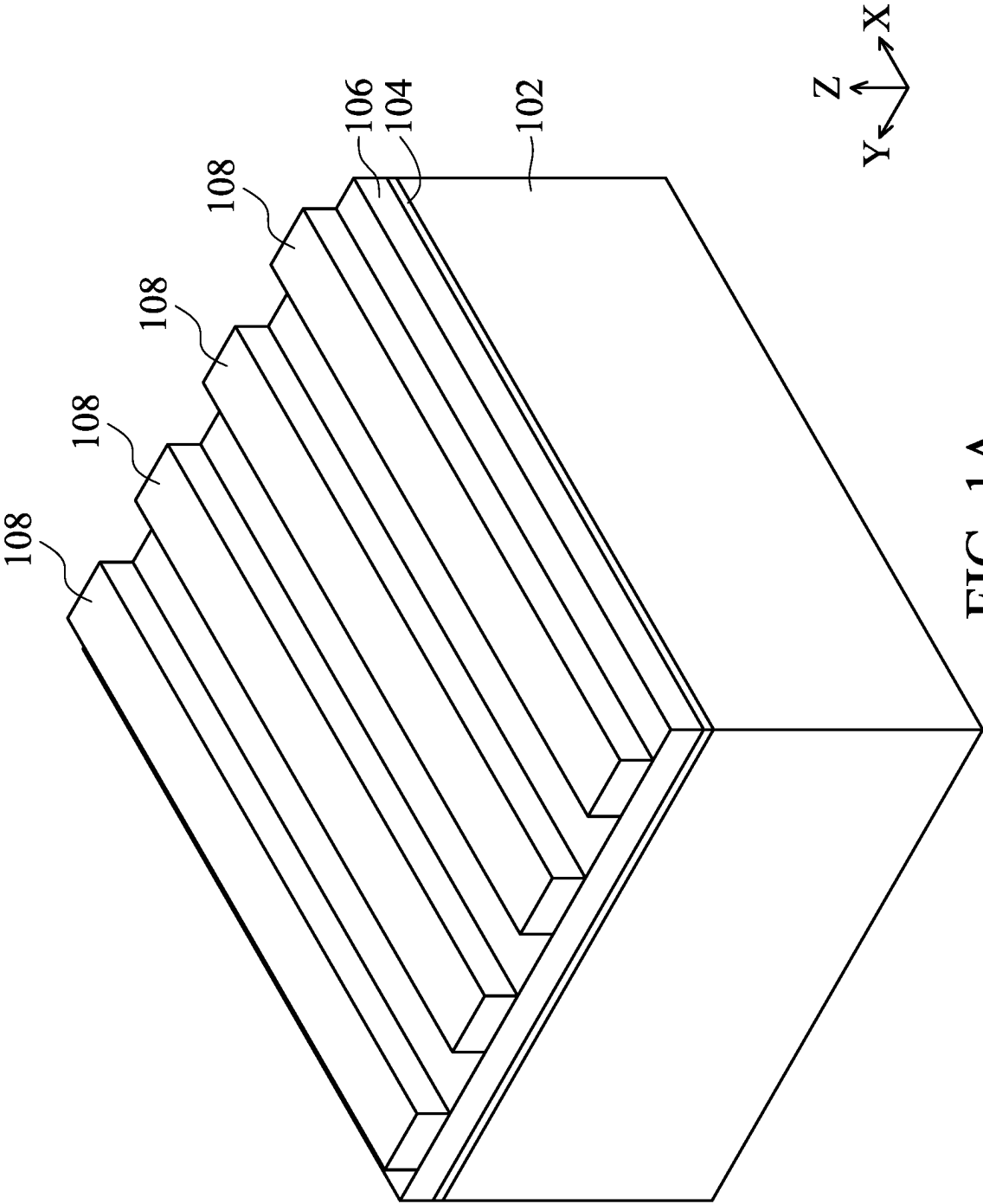
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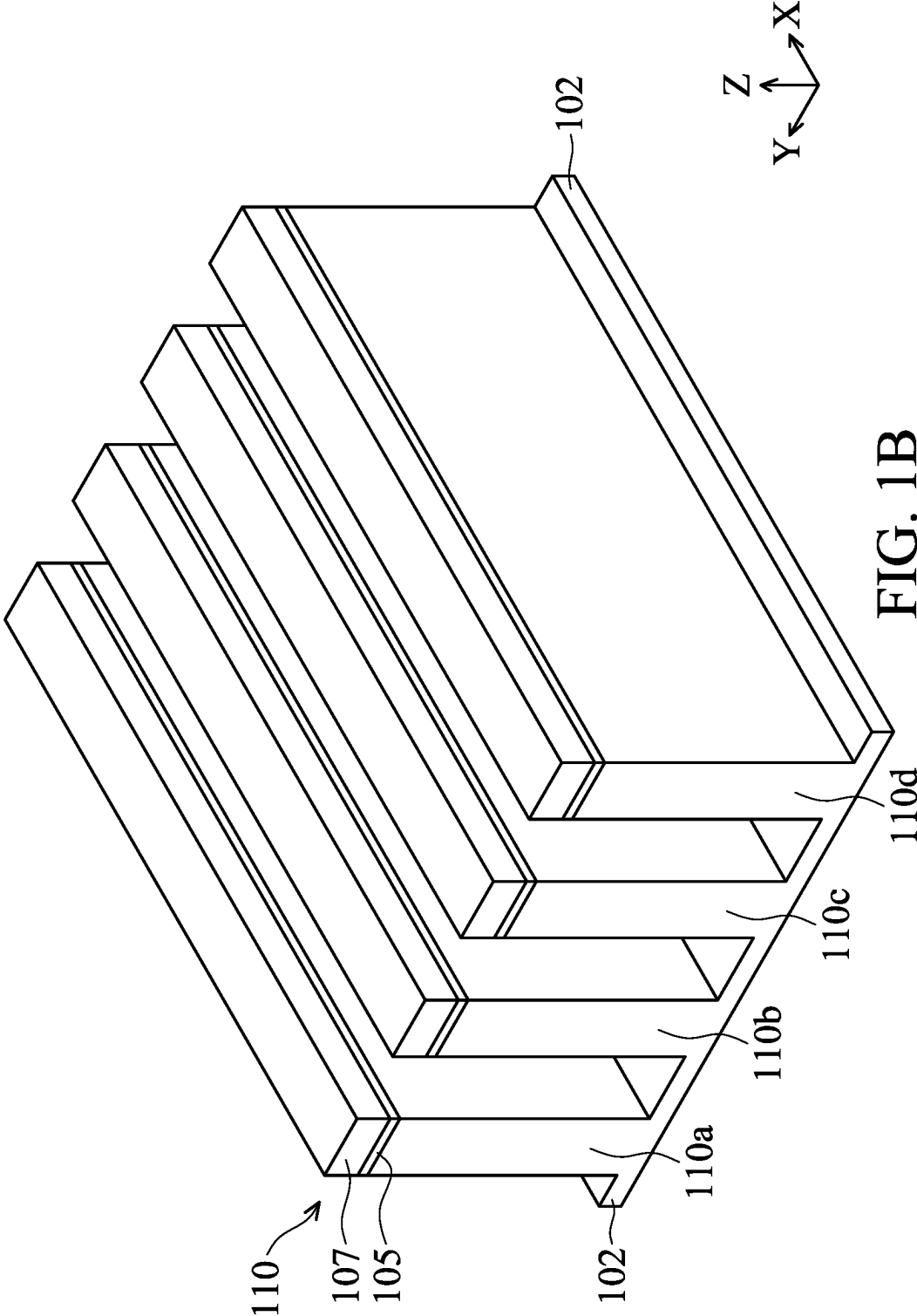
(57) **ABSTRACT**

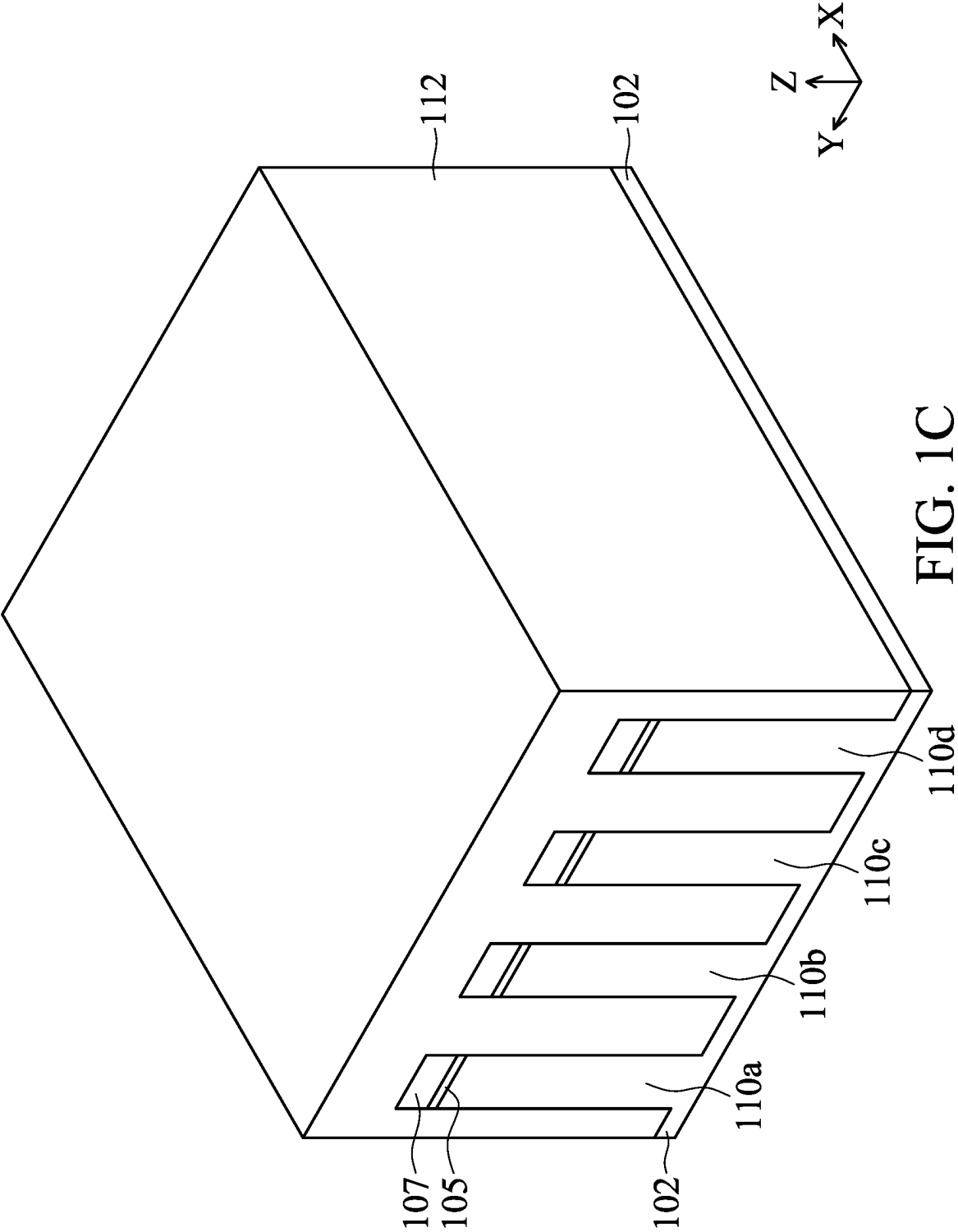
A semiconductor structure is provided, and includes a first fin structure, a second fin structure, and a third fin structure over a substrate. The second fin structure is located between the first fin structure and the third fin structure. The semiconductor structure also includes a fin isolation structure formed between the first fin structure and the third fin structure; and a gate structure formed over the first fin structure, the second fin structure, the third fin structure and the fin isolation structure. The semiconductor structure further includes a plurality of epitaxial structures formed over the first fin structure, the second fin structure and the third fin structure. The semiconductor structure includes a dielectric material over the first epitaxial structure, the second epitaxial structure, and the third epitaxial structure; and a contact formed in the dielectric material and connected to the first epitaxial structure and the third epitaxial structure.

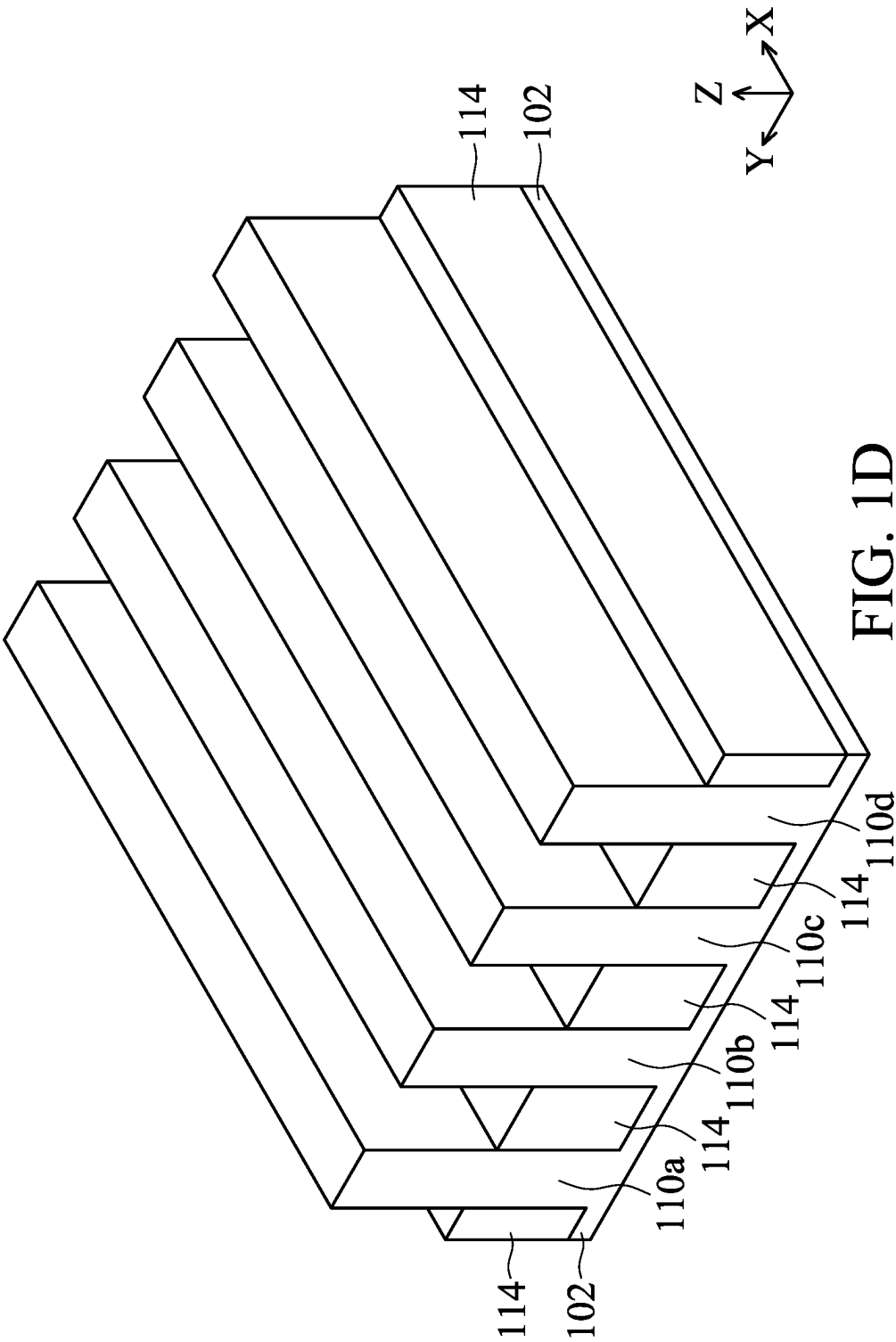
20 Claims, 21 Drawing Sheets

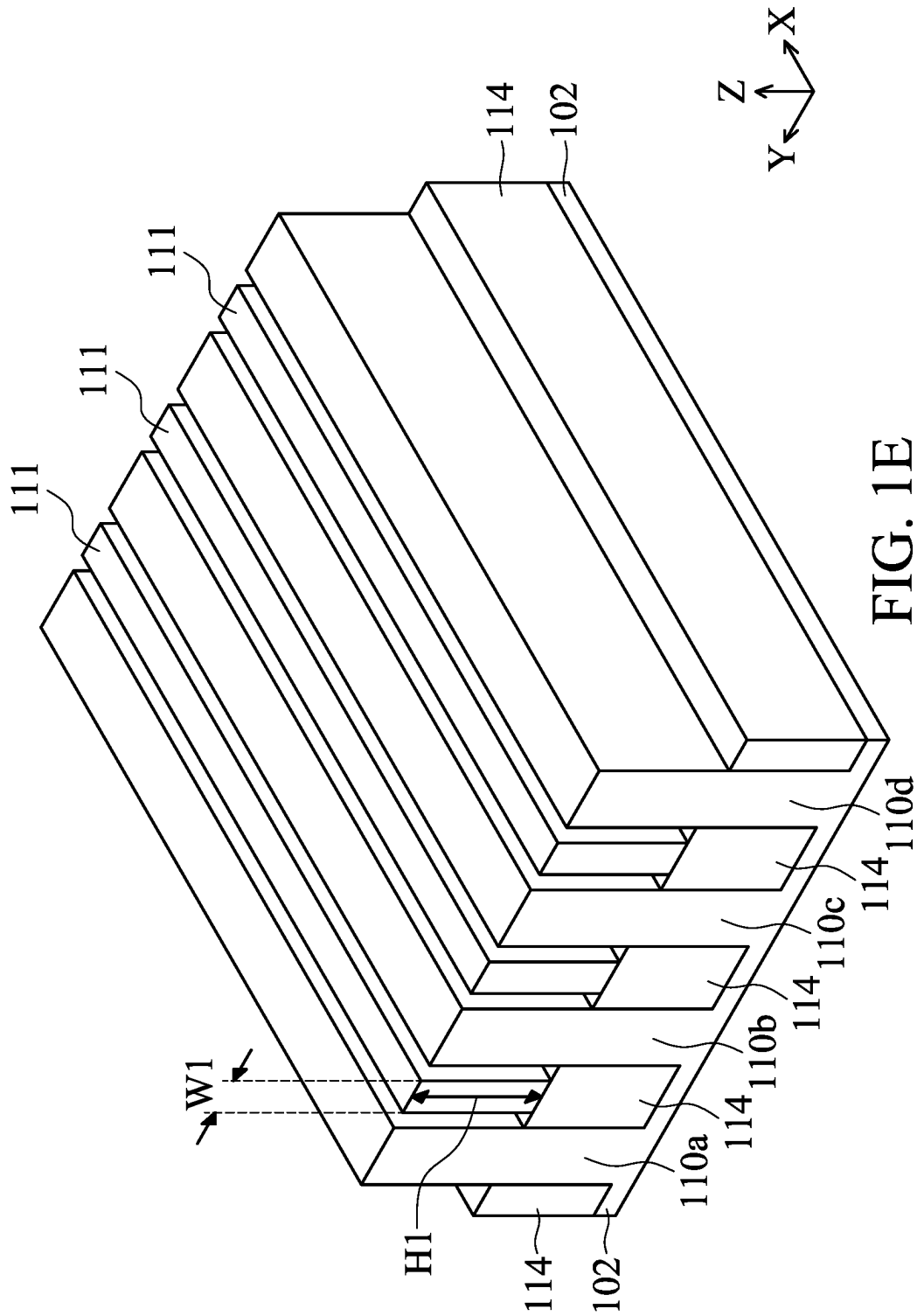


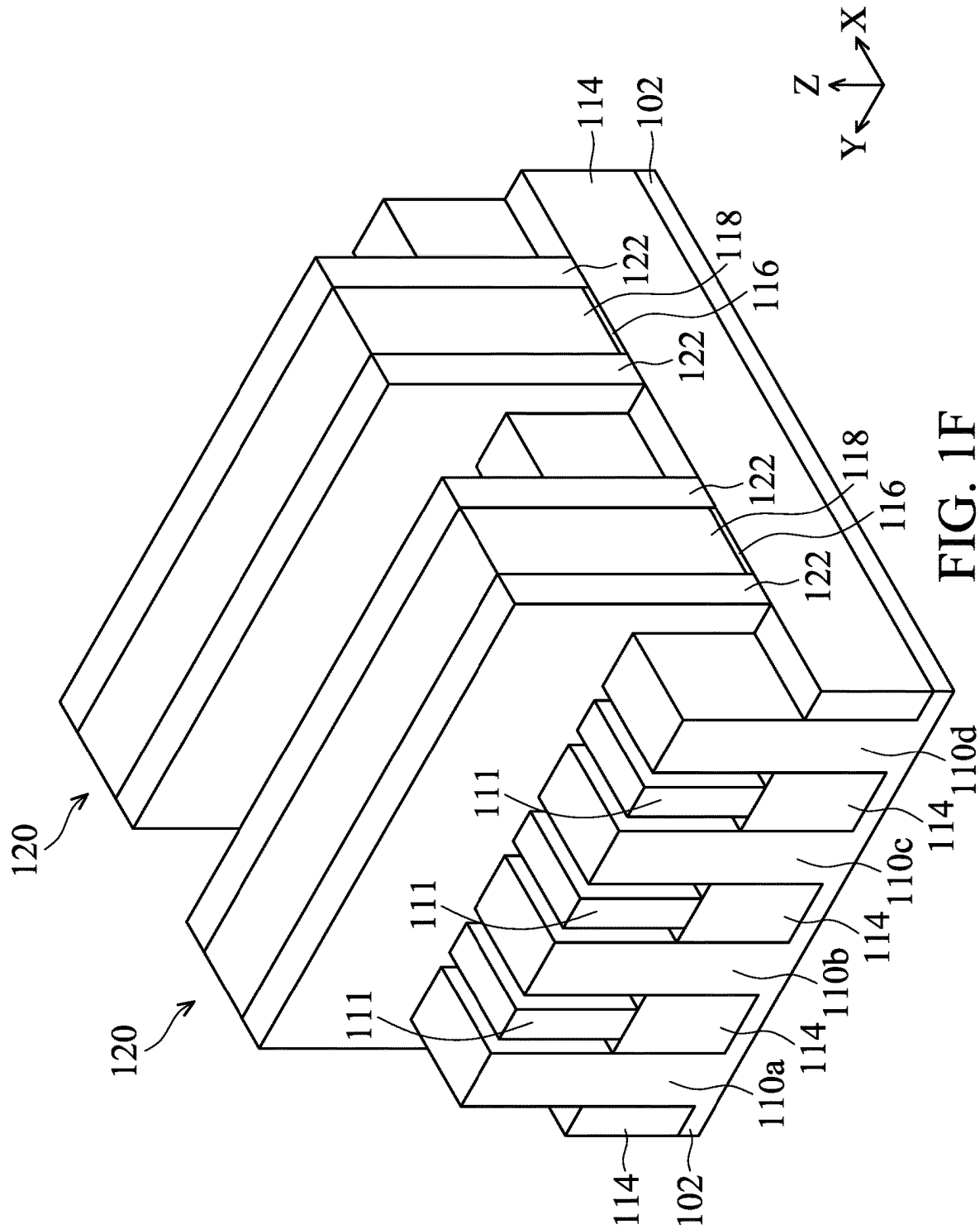


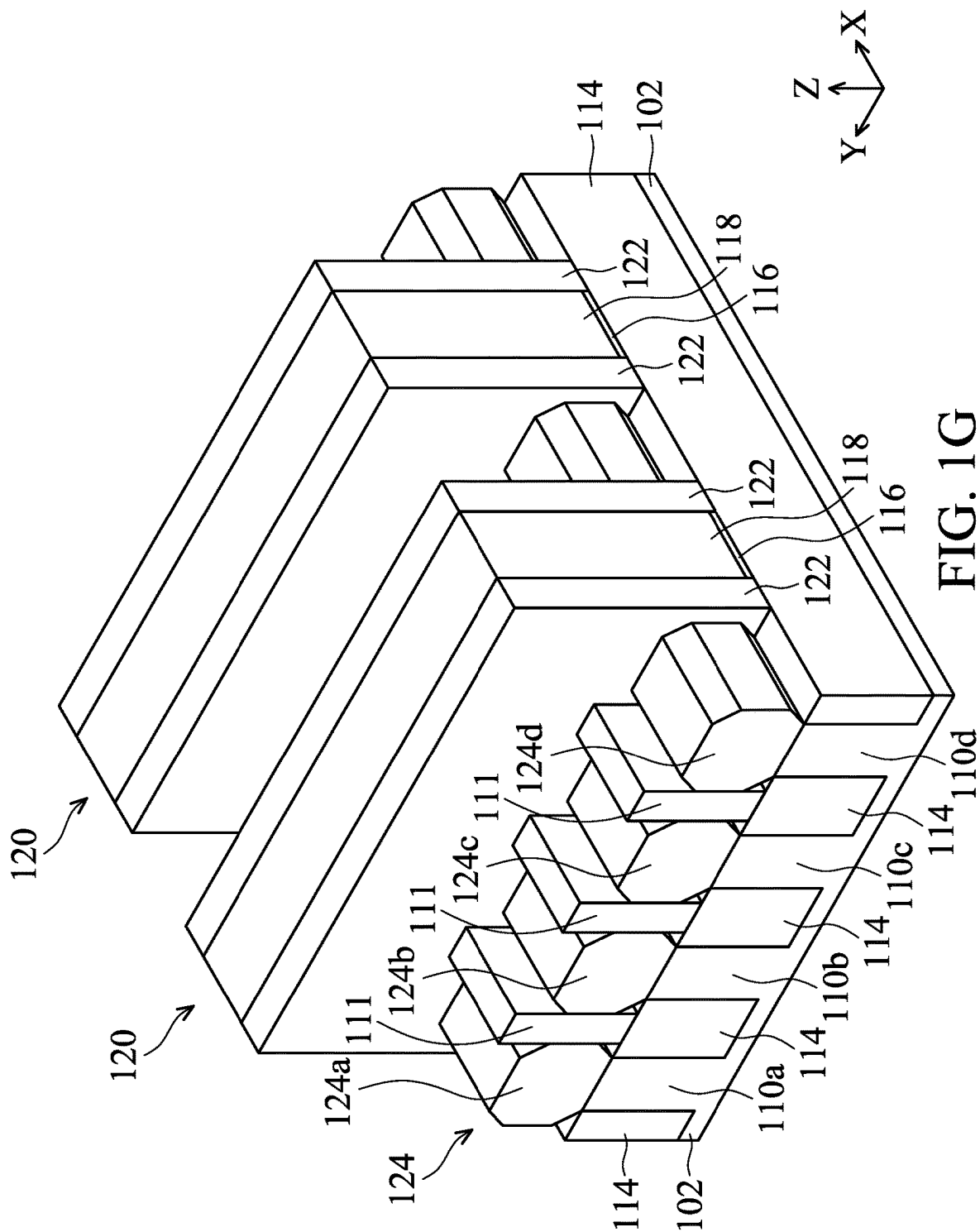


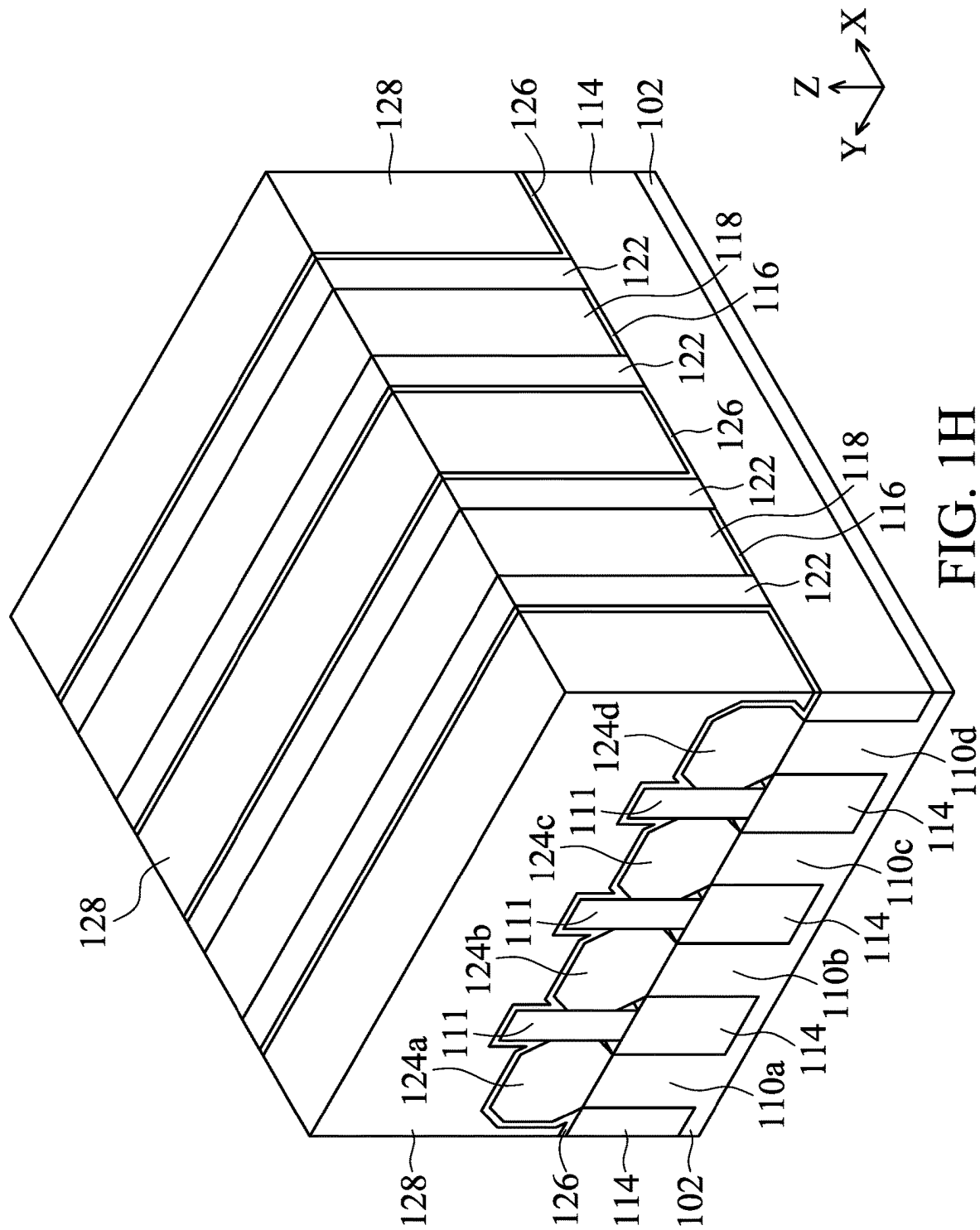


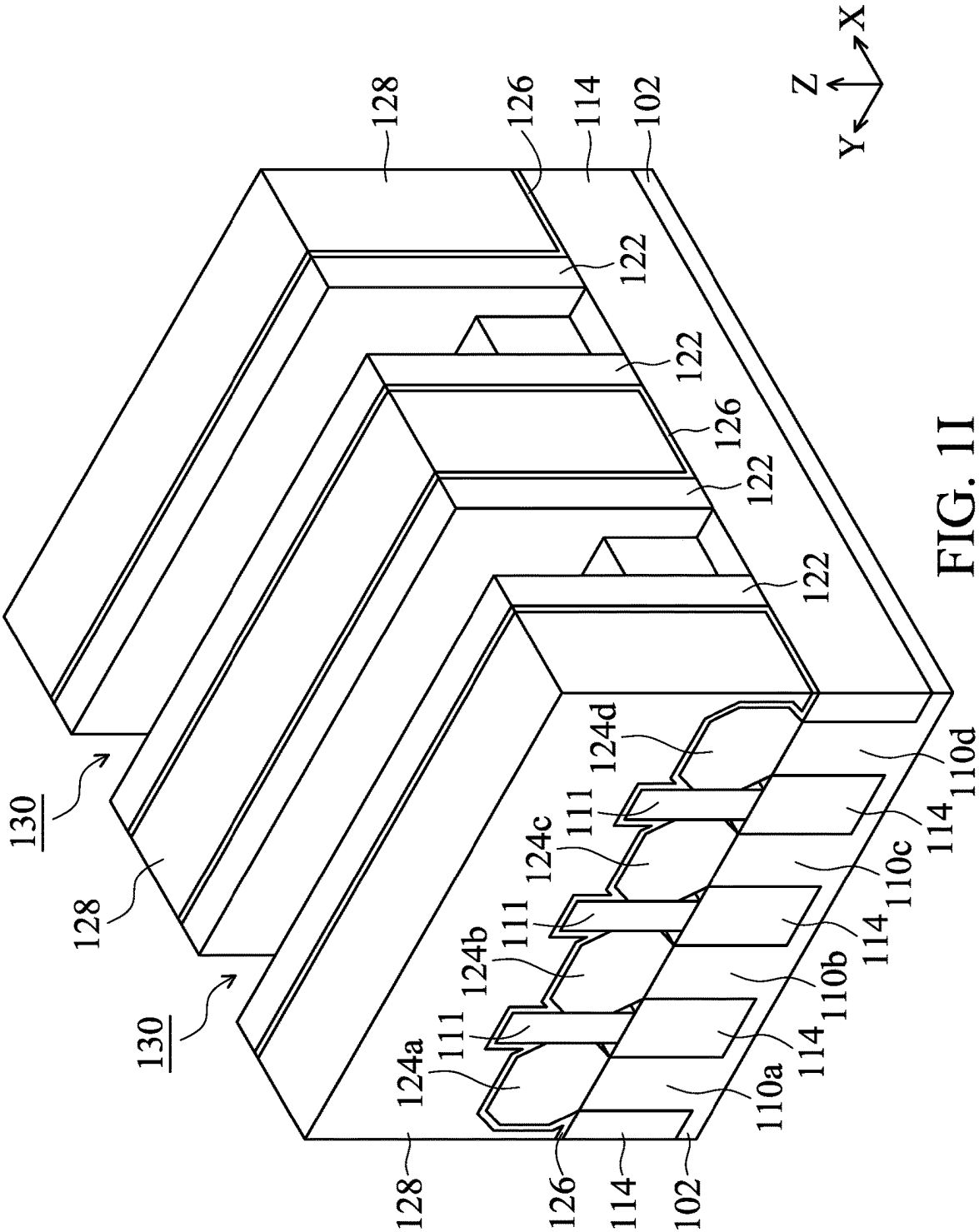












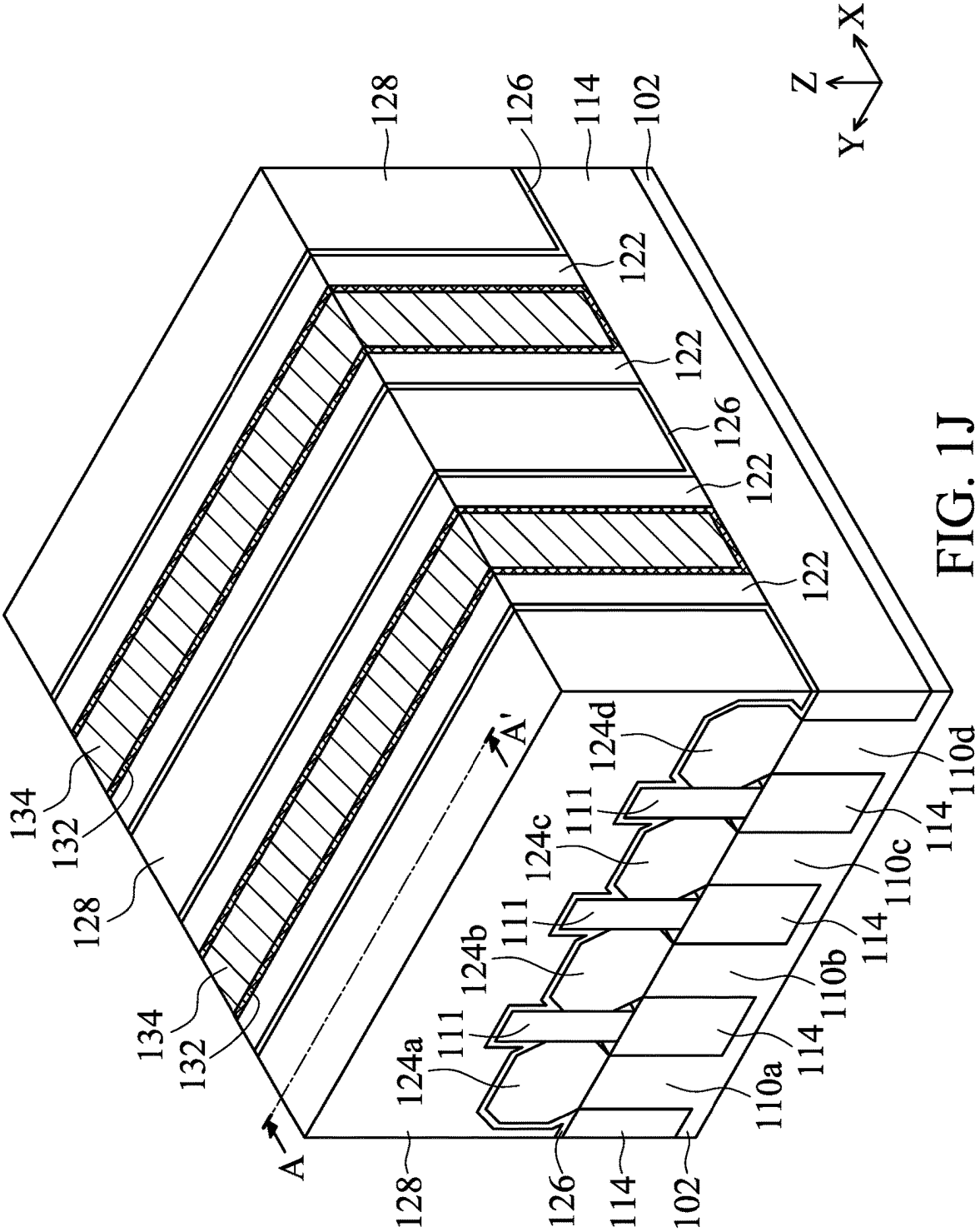


FIG. 1J

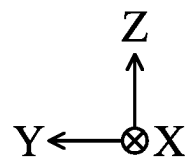
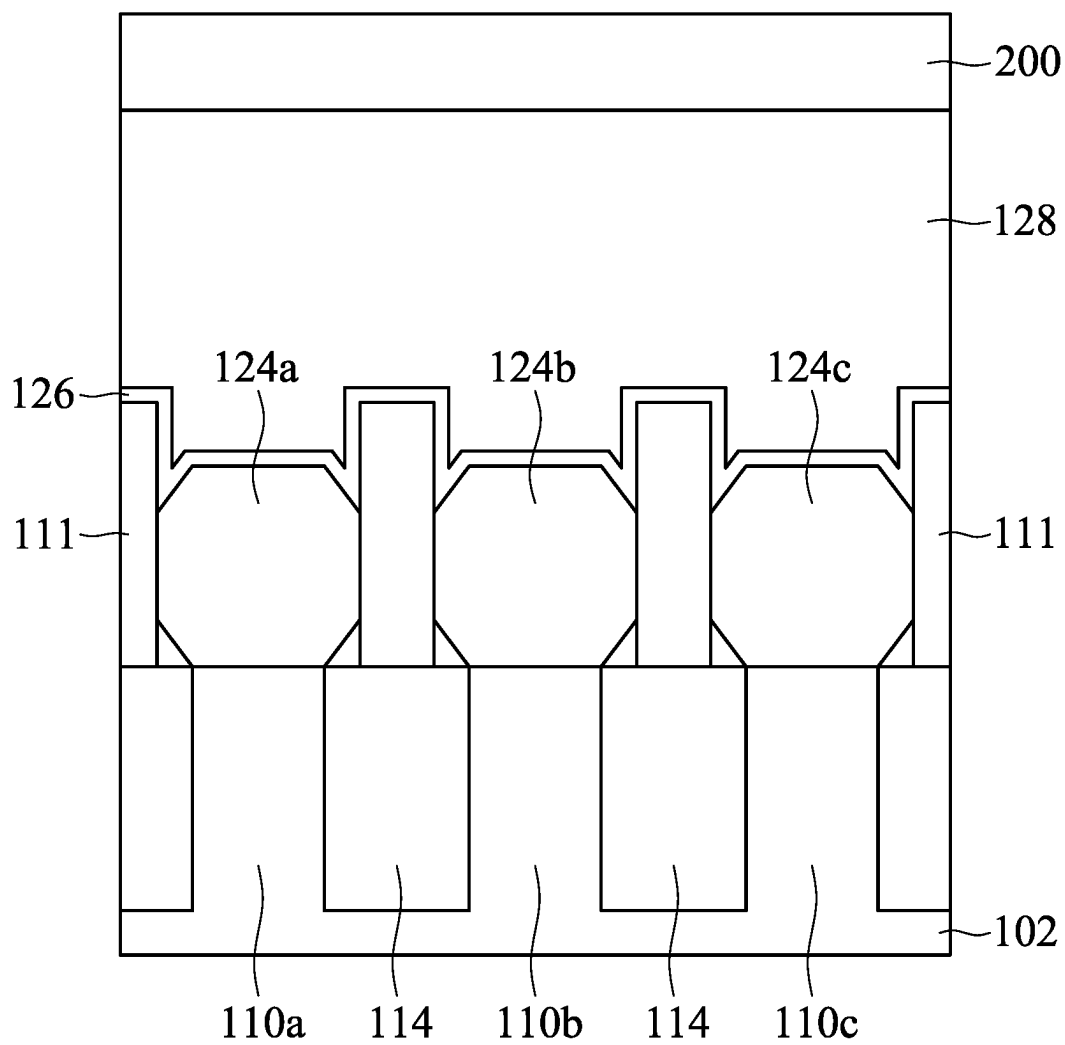


FIG. 2A

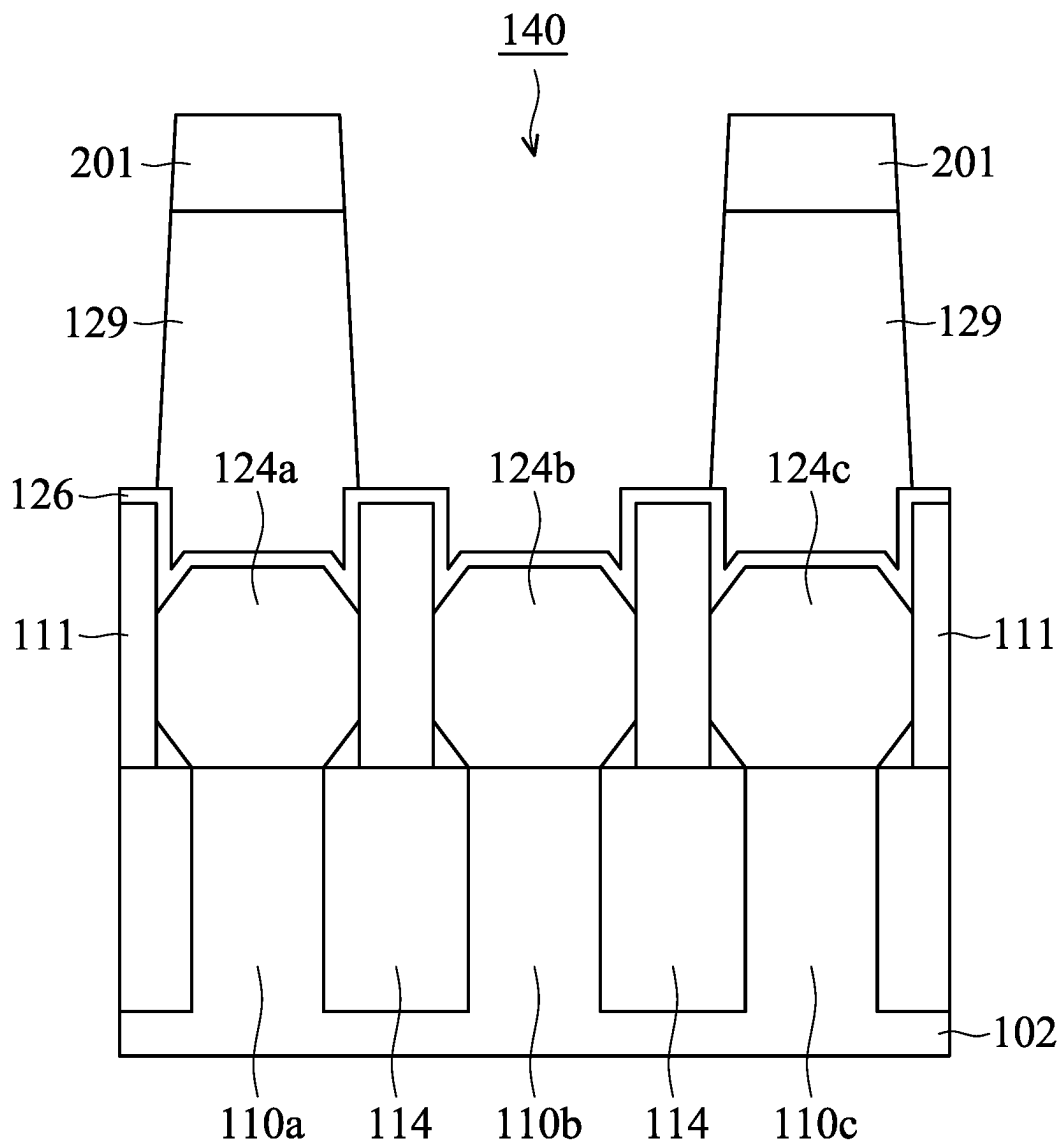


FIG. 2B

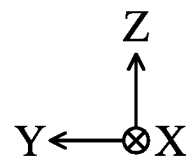
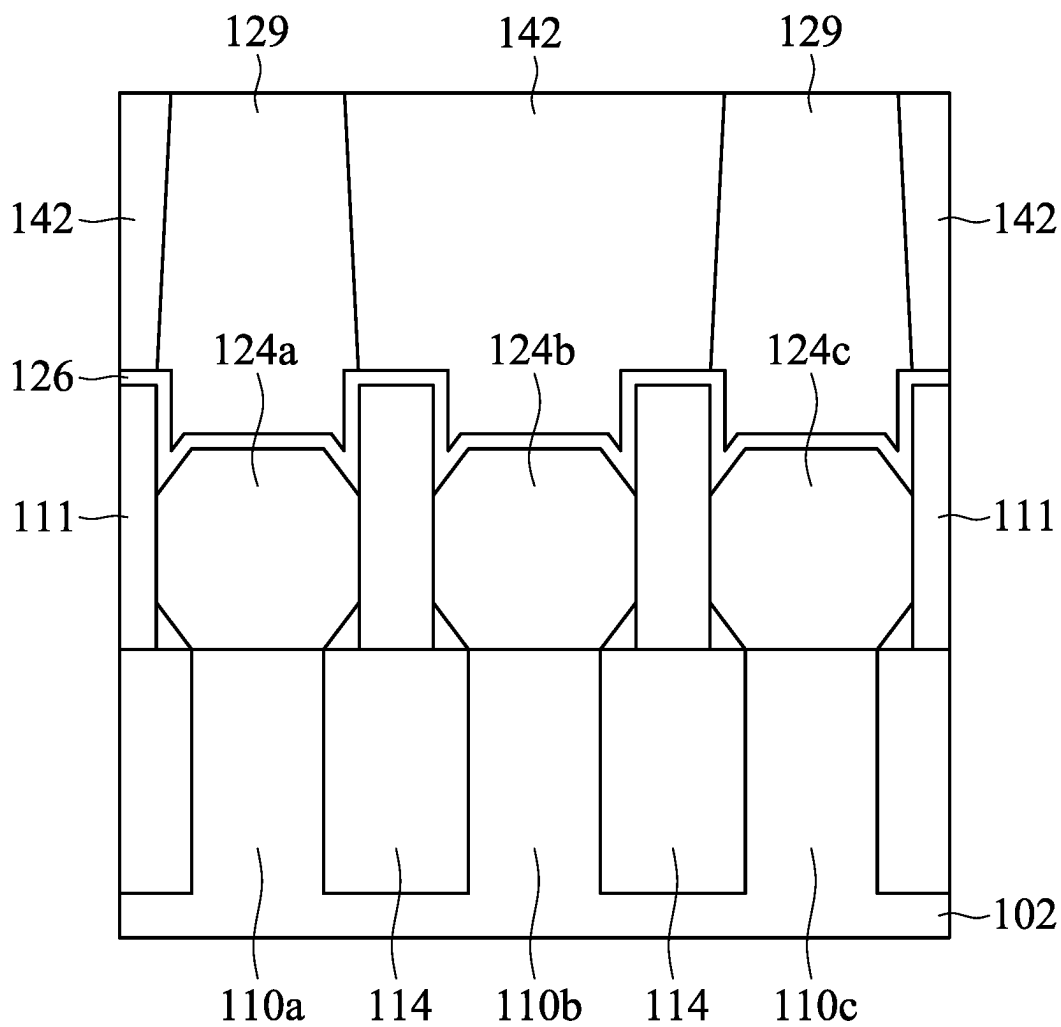


FIG. 2C

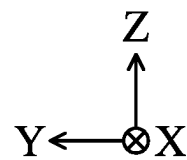
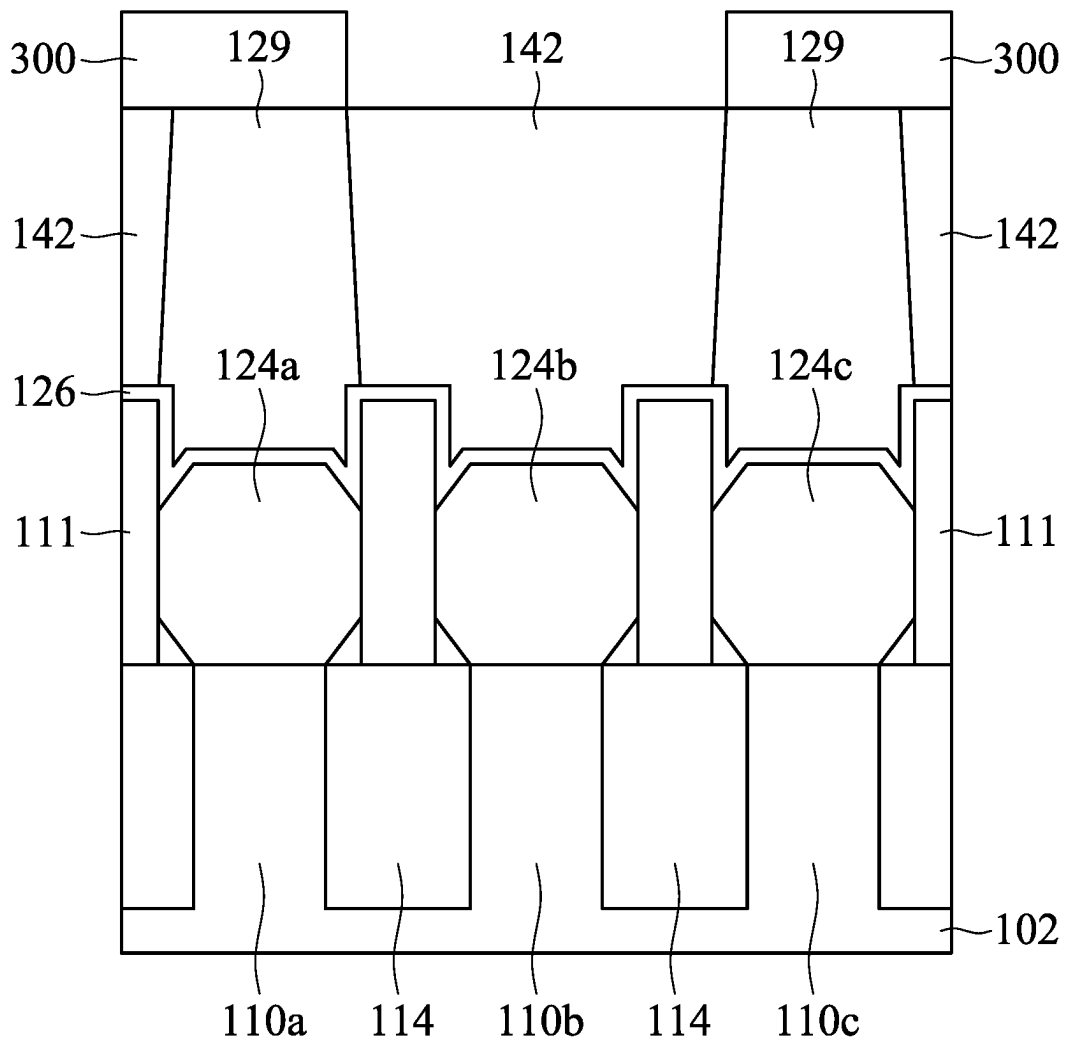


FIG. 2D

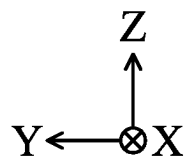
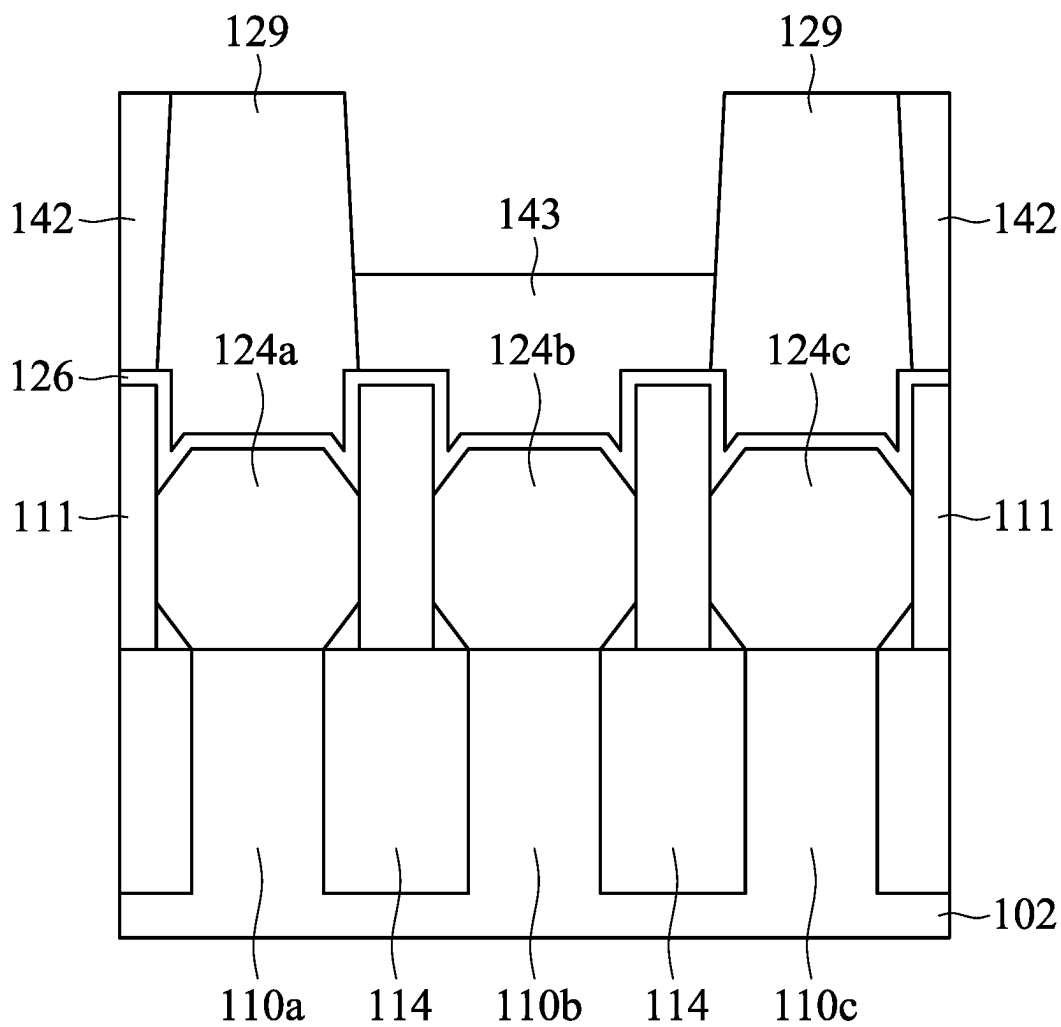


FIG. 2E

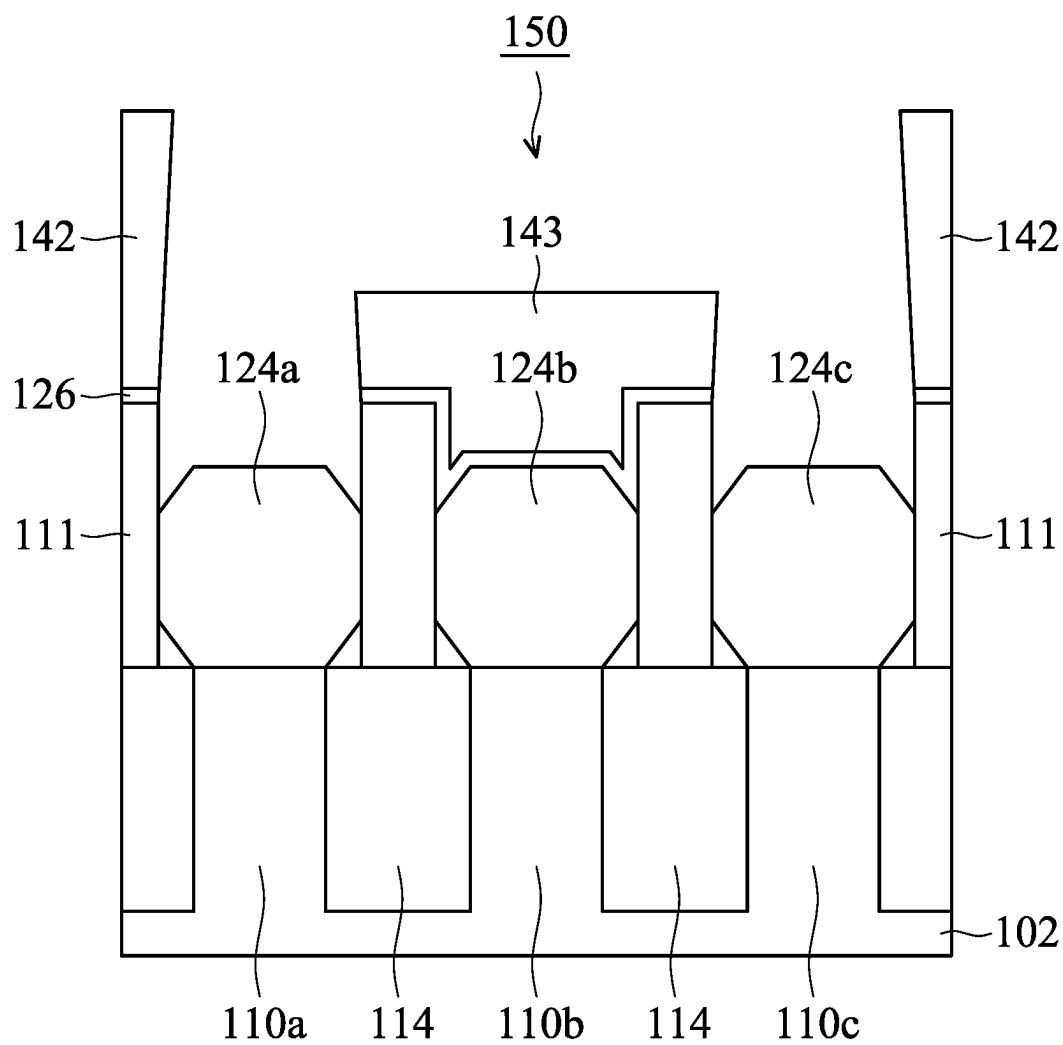


FIG. 2F

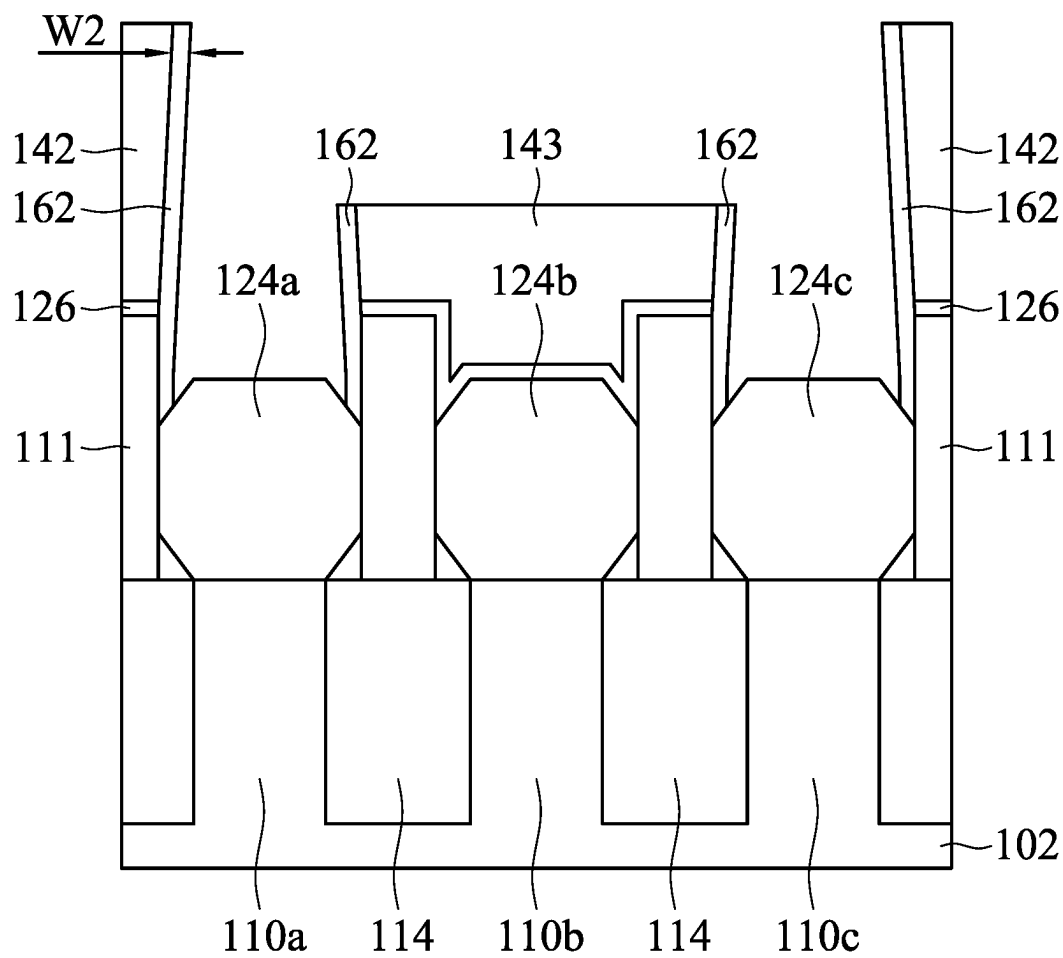


FIG. 2G

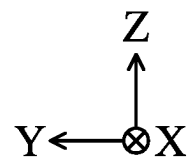
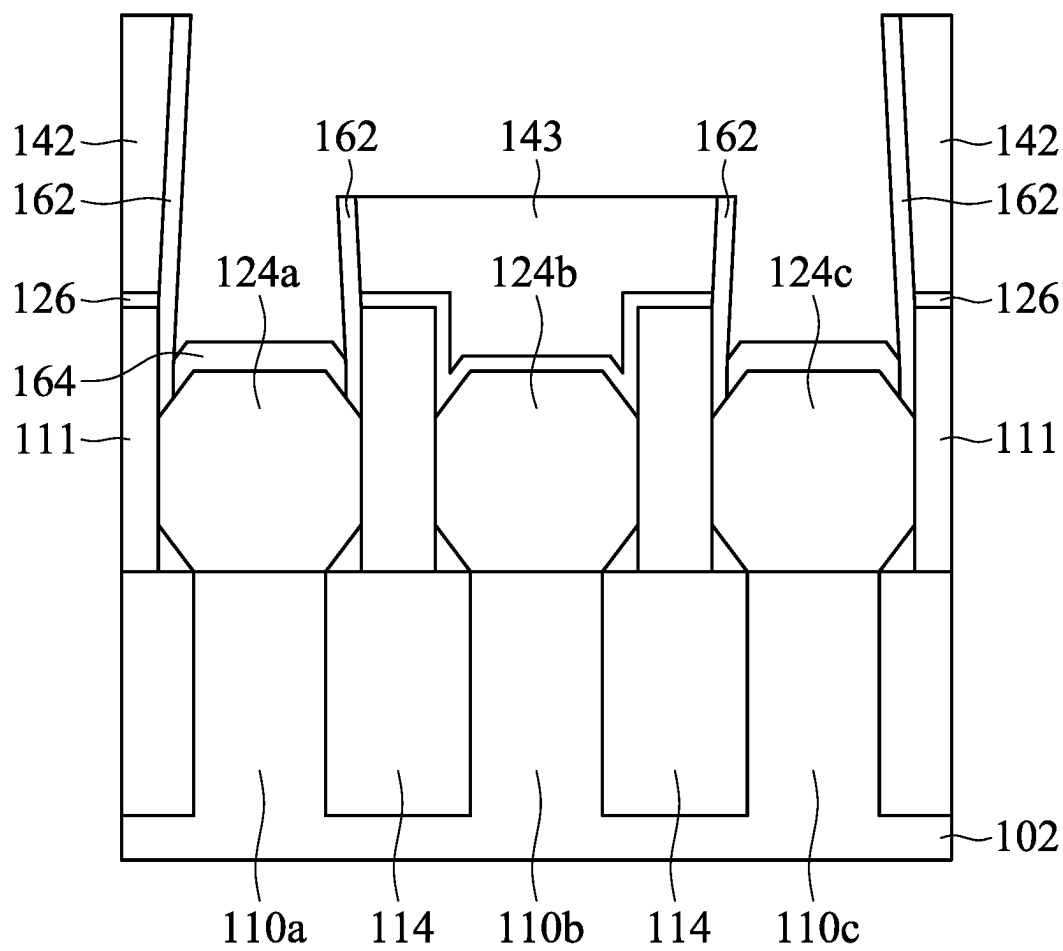


FIG. 2H

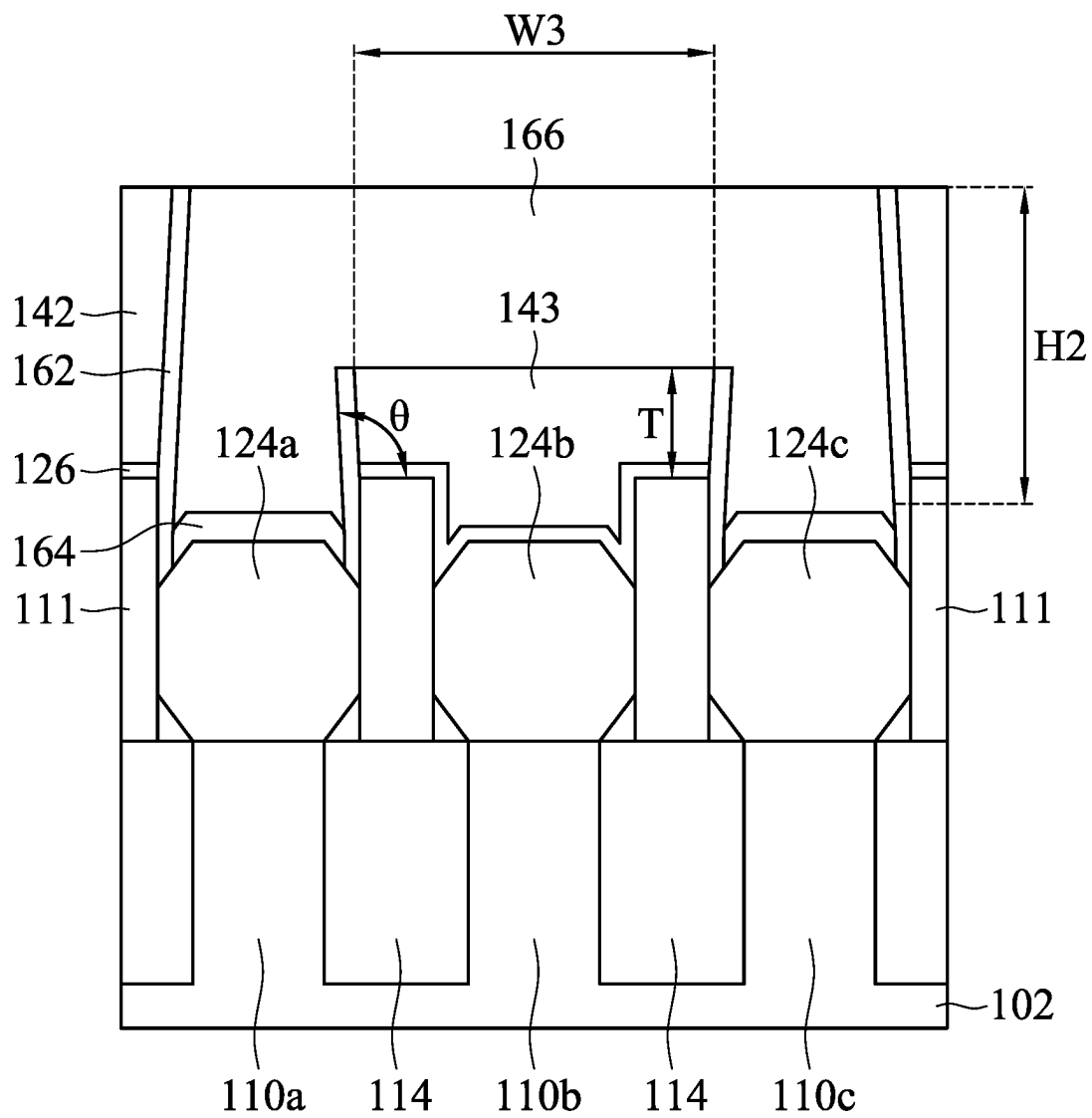


FIG. 2I

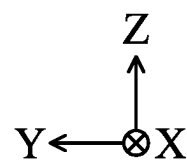
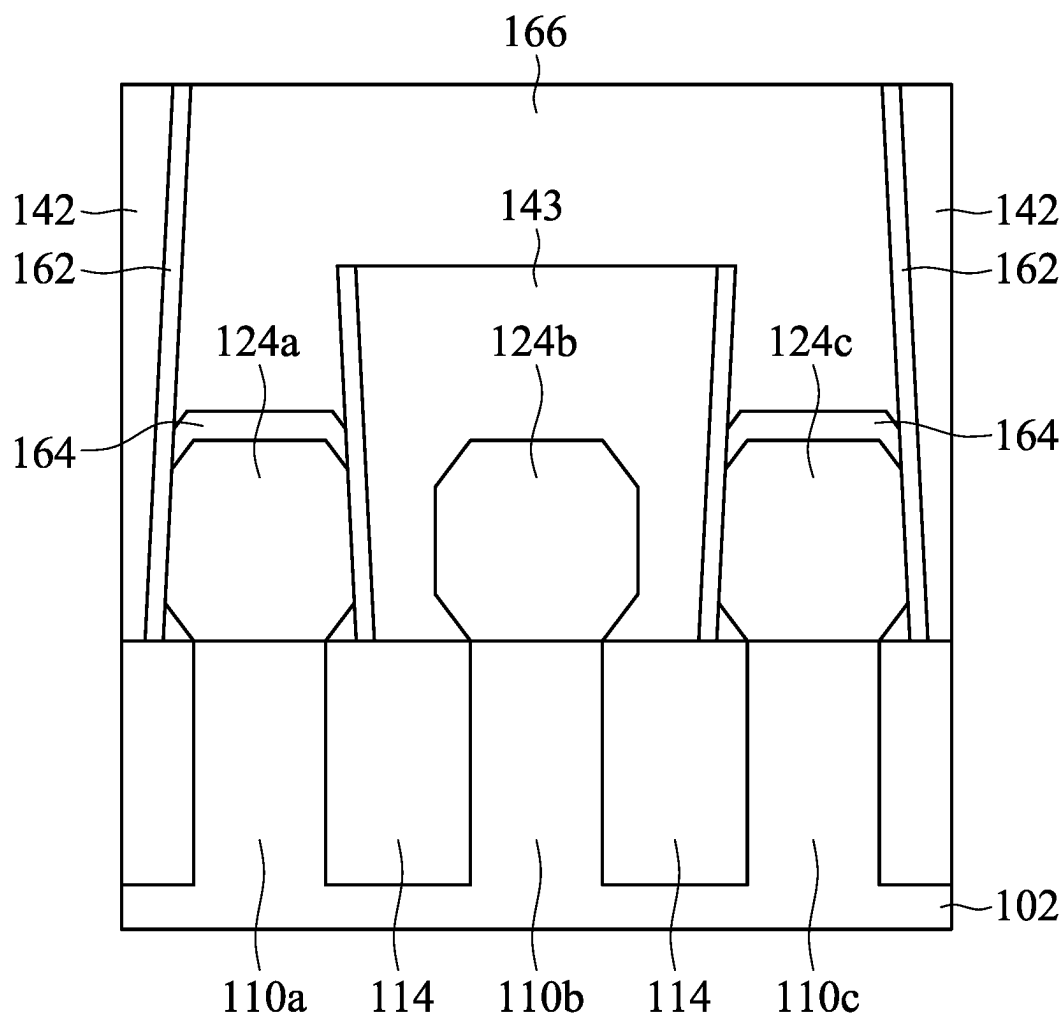


FIG. 3

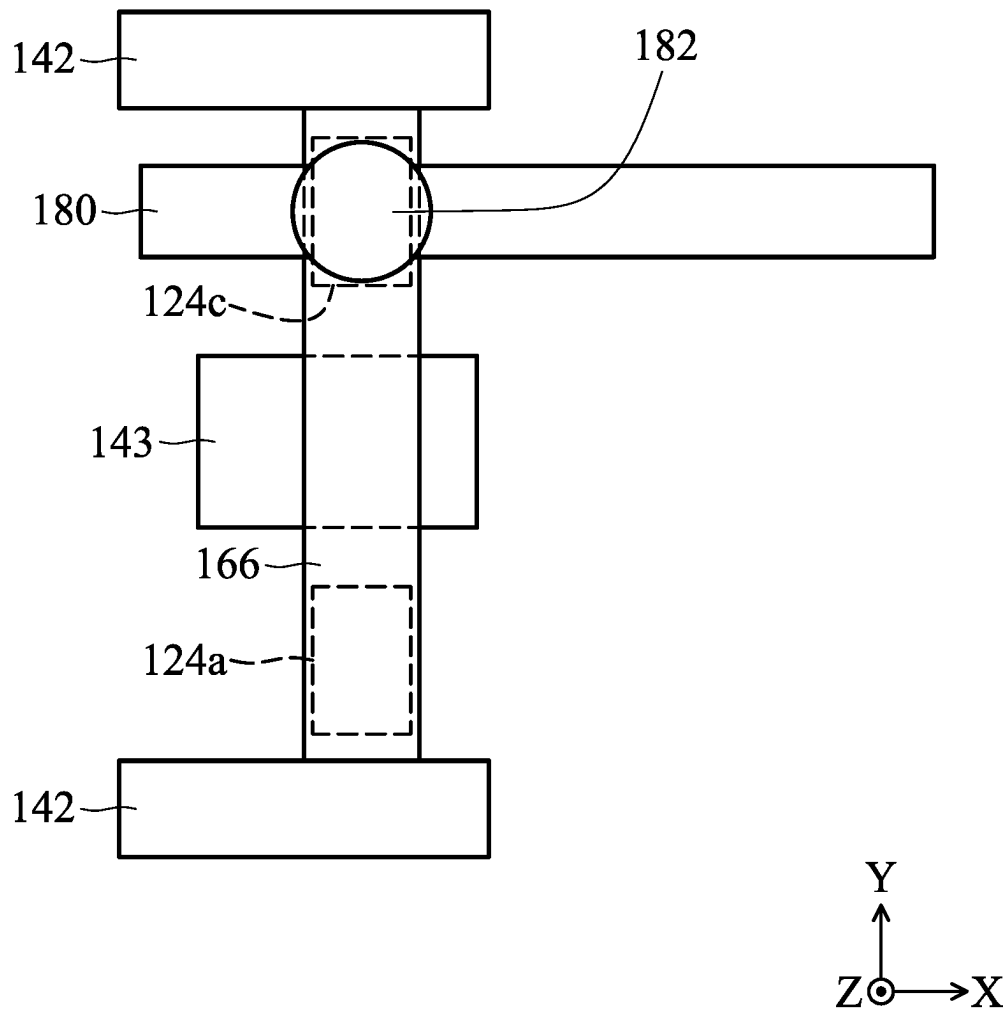


FIG. 4

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SEMICONDUCTOR STRUCTURE AND METHOD FOR FORMING THE SAME

BACKGROUND

Semiconductor devices are used in a variety of electronic applications, such as personal computers, cell phones, digital cameras, and other electronic equipment. Semiconductor devices are typically fabricated by sequentially depositing insulating or dielectric layers, conductive layers, and semi-conductive layers of material over a semiconductor substrate, and patterning the various material layers using lithography to form circuit components and elements thereon. Many integrated circuits are typically manufactured on a single semiconductor wafer, and individual dies on the wafer are singulated by sawing between the integrated circuits along a scribe line. The individual dies are typically packaged separately, in multi-chip modules, for example, or in other types of packaging.

As the semiconductor industry has progressed into nanometer technology process nodes in pursuit of higher device density, higher performance, and lower costs, challenges from both fabrication and design issues have resulted in the development of three-dimensional designs, such as the fin field effect transistor (FinFET). FinFETs are fabricated with a thin vertical “fin” (or fin structure) extending from the substrate. The channel of the FinFET is formed in this vertical fin. A gate is provided over the fin. The advantages of a FinFET may include reducing the short channel effect and providing a higher current flow.

Although existing FinFET devices and methods of fabricating FinFET devices have generally been adequate for their intended purposes, they have not been entirely satisfactory in all respects.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIGS. 1A-1J are perspective views illustrating various stages of forming a semiconductor structure in accordance with some embodiments.

FIGS. 2A-2I are cross-sectional views illustrating various stages of forming the semiconductor structure in accordance with some embodiments.

FIG. 3 is a cross-sectional view illustrating the semiconductor structure in accordance with some embodiments.

FIG. 4 is a layout view illustrating the semiconductor structure in accordance with some embodiments.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which addi-

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tional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

Fin structures described below may be patterned by any suitable method. For example, the fin structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the fin structures.

Embodiments of semiconductor structures and methods for forming the same are provided. The semiconductor structure may include a contact formed in the single layer and connected to the non-adjacent epitaxial structures. Accordingly, it is more likely to achieve miniaturization (for example, reducing cell height), and increase the routing flexibility. In addition, the formation of the contact helps to keep the remaining structure from being removed by the etching processes since different processes and materials are adopted.

FIGS. 1A-1J are perspective views illustrating various stages of forming a semiconductor structure in accordance with some embodiments. A substrate **102** is provided, as shown in FIG. 1A in accordance with some embodiments. The substrate **102** may be a semiconductor wafer such as a silicon wafer. Alternatively or additionally, the substrate **102** may include elementary semiconductor materials, compound semiconductor materials, and/or alloy semiconductor materials. Examples of the elementary semiconductor materials may include, but are not limited to, crystal silicon, polycrystalline silicon, amorphous silicon, germanium, and/or diamond. Examples of the compound semiconductor materials may include, but are not limited to, silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide. Examples of the alloy semiconductor materials may include, but are not limited to, SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP. In some embodiments, the substrate **102** includes an epitaxial layer. For example, the substrate **102** has an epitaxial layer overlying a bulk semiconductor.

Afterwards, a dielectric layer **104** and a mask layer **106** are formed over the substrate **102**, and a patterned photoresist layer **108** is formed over the mask layer **106**, as shown in FIG. 1A in accordance with some embodiments. The

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patterned photoresist layer **108** may be formed by a deposition process and a patterning process.

The deposition process for forming the patterned photoresist layer **108** may include a chemical vapor deposition (CVD) process, a high-density plasma chemical vapor deposition (HDPCVD) process, a spin-on process, a sputtering process, or another applicable process. The patterning process for forming the patterned photoresist layer **108** may include a photolithography process and an etching process. The photolithography process may include photoresist coating (e.g., spin-on coating), soft baking, mask aligning, exposure, post-exposure baking, developing the photoresist, rinsing, and drying (e.g., hard baking). The etching process may include a dry etching process or a wet etching process.

Moreover, the dielectric layer **104** may be a buffer layer between the substrate **102** and the mask layer **106**. In some embodiments, the dielectric layer **104** is used as a stop layer when the mask layer **106** is removed. The dielectric layer **104** may be made of silicon oxide. The mask layer **106** may be made of silicon oxide, silicon nitride, silicon oxynitride, or another applicable material. In some other embodiments, more than one mask layer **106** is formed over the dielectric layer **104**.

The dielectric layer **104** and the mask layer **106** may be formed by deposition processes, which may include a chemical vapor deposition (CVD) process, a high-density plasma chemical vapor deposition (HDPCVD) process, a spin-on process, a sputtering process, or another applicable process.

After the patterned photoresist layer **108** is formed, the dielectric layer **104** and the mask layer **106** are patterned by using the patterned photoresist layer **108** as a mask, as shown in FIG. 1B in accordance with some embodiments. As a result, a patterned dielectric layer **105** and a patterned mask layer **107** are obtained. Afterwards, the patterned photoresist layer **108** is removed.

Next, an etching process is performed on the substrate **102** to form a plurality of fin structures (for example, including a first fin structure **110a**, a second fin structure **110b**, a third fin structure **110c**, a fourth fin structure **110d**, which may be collectively referred to as the fin structures **110a**, **110b**, **110c**, and **110d**) by using the patterned dielectric layer **105** and the patterned mask layer **107** as a mask. The etching process may be a dry etching process or a wet etching process.

In some embodiments, the substrate **102** is etched by a dry etching process. The dry etching process includes using a fluorine-based etchant gas, such as SF_6 , C_xF_y , NF_3 or a combination thereof. The etching process may be a time-controlled process, and continue until the fin structures **110a**, **110b**, **110c**, and **110d** reach a predetermined height. In some other embodiments, the fin structures **110a**, **110b**, **110c**, and **110d** have a width that gradually increases from the top portion to the lower portion.

After the fin structures **110a**, **110b**, **110c**, and **110d** are formed, an insulating layer **112** is formed to cover the fin structures **110a**, **110b**, **110c**, and **110d**, the patterned pad layer **105**, and the patterned mask layer **107** over the substrate **102**, as shown in FIG. 1C in accordance with some embodiments.

In some embodiments, the insulating layer **112** is made of silicon oxide, silicon nitride, silicon oxynitride, fluoride-doped silicate glass (FSG), or another low-k dielectric material. The insulating layer **112** may be deposited using a chemical vapor deposition (CVD) process, a spin-on-glass process, or another applicable process.

Next, the insulating layer **112** is thinned or planarized to expose the top surface of the patterned mask layer **107**. In

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some embodiments, the insulating layer **112** is thinned by a chemical mechanical polishing (CMP) process. Afterwards, the patterned dielectric layer **105** and the patterned mask layer **107** are removed.

After the patterned dielectric layer **105** and the patterned mask layer **107** are removed, an upper portion of the insulating layer **112** is removed to form an isolation structure **114**, as shown in FIG. 1D in accordance with some embodiments. The isolation structure **114** may be a shallow trench isolation (STI) structure surrounding the fin structures **110a**, **110b**, **110c**, and **110d**. That is to say, the isolation structure **114** is located between any adjacent two of the fin structures **110a**, **110b**, **110c**, and **110d**.

In some embodiments, a portion of the fin structures **110a**, **110b**, **110c**, and **110d** is embedded in the isolation structure **114**. More specifically, a lower portion of the fin structures **110a**, **110b**, **110c**, and **110d** is surrounded by the isolation structure **114**, while an upper portion of the fin structures **110a**, **110b**, **110c**, and **110d** protrudes from the isolation structure **114**. The isolation structure **114** is configured to prevent electrical interference or crosstalk.

After the isolation structure **114** is formed, fin isolation structures **111** are formed over the isolation structure **114**, as shown in FIG. 1E in accordance with some embodiments. Each of the fin isolation structures **111** has a strip shape that is similar to the shape of each of the fin structures **103**. Therefore, the fin isolation structures **111** may be referred to as dummy fin structures or hybrid fin structures, and the fin structures **110a**, **110b**, **110c**, and **110d** may be referred to as active fin structures.

The fin isolation structures **111** are separated from the substrate **102** by the isolation structure **114** and extend along a direction that is substantially parallel to the longitudinal direction of the fin structures **110a**, **110b**, **110c**, and **110d**. In some embodiments, the top surface of the fin isolation structures **111** is substantially level with the top surface of the fin structures **110a**, **110b**, **110c**, and **110d**. In some embodiments, the patterned dielectric layer **105** and the patterned mask layer **107** are not removed, and the distance between the top surface of the patterned mask layer **107** to the top surface of the isolation structure **114** is substantially the same as the distance between the top surface and the bottom surface of the fin isolation structures **111** (i.e. the height **H1** of the fin isolation structures **111**).

In some embodiments, the width **W1** of the fin isolation structures **111** is about 10 nm to about 50 nm, and the height **H1** of the fin isolation structures **111** is about 5 nm to about 40 nm. As such, the fin isolation structures **111** may provide proper insulation for the subsequently formed epitaxial structures (for example, shown in FIG. 1G). However, the present disclosure is not limited thereto. In some embodiments, the fin isolation structures **111** may be omitted.

In some embodiments, some of the fin isolation structures **111** serve as portions of insulating gate-cut structures and are made of a nitride-based material, such as silicon nitride, silicon oxynitride, or silicon carbon nitride, or the like. In some embodiments, each of the fin isolation structures **111** includes a single-layered structure. In some other embodiments, each of the fin isolation structures **111** includes a double-layered structure, which is made of two layers having different materials. In some embodiments, the fin isolation structures **111** are made of a high-k dielectric material, such as metal oxide. Examples of high-k dielectric materials include hafnium oxide (HfO_2), hafnium silicon oxide (Hf-SiO), hafnium tantalum oxide (HfTaO), hafnium titanium oxide (HfTiO), hafnium zirconium oxide (HfZrO), zirconium oxide, titanium oxide, aluminum oxide, or other appli-

cable dielectric materials. In some embodiments, the fin isolation structures **111** are formed by a chemical vapor deposition (CVD) process, a physical vapor deposition process (PVD), an atomic layer deposition (ALD) process, or another applicable process.

After the fin isolation structures **111** are formed, dummy gate structures **120** are formed across the fin structures **110a**, **110b**, **110c**, and **110d**, the fin isolation structures **111**, and extend over the isolation structure **114**, as shown in FIG. 1F in accordance with some embodiments. In some embodiments, each of the dummy gate structures **120** includes a dummy gate dielectric layer **116** and a dummy gate electrode layer **118** formed over the dummy gate dielectric layer **116**. After the dummy gate structures **120** are formed, gate spacers **122** are formed on opposite sidewalls of each of the dummy gate structures **120**. Each of the gate spacers **122** may be a single layer or multiple layers.

In order to improve the speed of the FinFET device structure, the gate spacers **122** are made of low-k dielectric materials. In some embodiments, the low-k dielectric materials have a dielectric constant (k value) less than about 4. Examples of low-k dielectric materials include, but are not limited to, silicon oxide, silicon nitride, silicon carbonitride (SiCN), silicon oxide carbonitride (SiOCN), fluorinated silica glass (FSG), carbon doped silicon oxide, amorphous fluorinated carbon, parylene, bis-benzocyclobutenes (BCB), or polyimide.

In some other embodiments, the gate spacers **122** are made of an extreme low-k (ELK) dielectric material with a dielectric constant (k-value) less than about 2.5. In some embodiments, the ELK dielectric materials include carbon doped silicon oxide, amorphous fluorinated carbon, parylene, bis-benzocyclobutenes (BCB), polytetrafluoroethylene (PTFE) (Teflon), or silicon oxycarbide polymers (SiOC). In some embodiments, ELK dielectric materials include a porous version of an existing dielectric material, such as hydrogen silsesquioxane (HSQ), porous methyl silsesquioxane (MSQ), porous polyarylether (PAE), porous SiLK, or porous silicon oxide (SiO₂).

In addition, in some embodiments, the gate spacers **122** include air gaps (not shown) to further reduce their k value, such that the capacitances between the gate structures (formed subsequently) and the contacts (formed subsequently) electrically connected to the epitaxial structure (formed subsequently) may be reduced.

Afterwards, epitaxial structures are formed over the fin structures **110a**, **110b**, **110c**, and **110d**, as shown in FIG. 1G in accordance with some embodiments. In some embodiments, a first epitaxial structure **124a** is formed over the first fin structure **110a**, a second epitaxial structure **124b** is formed over the second fin structure **110b**, a third epitaxial structure **124c** is formed over the third fin structure **110c**, and a fourth epitaxial structure **124d** is formed over the fourth fin structure **110d**. For the sake of simplicity, the first epitaxial structure **124a**, the second epitaxial structure **124b**, the third epitaxial structure **124c**, and the fourth epitaxial structure **124d** may be collectively referred to as the epitaxial structures **124**.

In some embodiments, the epitaxial structures **124** contact and are separated by the fin isolation structures **111**. In some embodiments, the top surface of the fin isolation structures **111** is higher than top surfaces of the epitaxial structures **124**. In some embodiments, the epitaxial structures **124** include different types of epitaxial structures. For example, the first epitaxial structure **124a** and the fourth epitaxial structure **124d** are the same type of epitaxial structure, and the second epitaxial structure **124b** and the third epitaxial

structure **124c** are another type of epitaxial structure that is different from the type of the first epitaxial structure **124a** and the fourth epitaxial structure **124d**. However, the configuration of the epitaxial structures **124** merely serves as an example, those skilled in the art would modify the types or shapes of the epitaxial structures **124** based on the present application.

In some embodiments, portions of the fin structure **110** adjacent to the dummy gate structures **120** are recessed to form recesses at two sides of the fin structure **110**, and a strained material is grown in the recesses by an epitaxial (epi) process to form the epitaxial structures **124**. In addition, the lattice constant of the strained material may be different from the lattice constant of the substrate **102**. In some embodiments, the epitaxial structures **124** include Ge, SiGe, InAs, InGaAs, InSb, GaAs, GaSb, InAlP, InP, or the like.

After the epitaxial structures **124** are formed, a contact etch stop layer (CESL) **126** is formed over the substrate **102**, and an inter-layer dielectric (ILD) layer **128** is formed over the CESL **126**. More specifically, the CESL **126** is formed over the epitaxial structures **124**, the fin isolation structures **111**, and the sidewalls of the gate spacers **122**. In some embodiments, the CESL **126** is made of silicon nitride, silicon oxynitride, and/or other applicable materials. Moreover, the CESL **126** may be formed by plasma enhanced CVD, low-pressure CVD, atomic layer deposition (ALD), or other applicable processes.

In some embodiments, the ILD layer **128** includes multilayers made of multiple dielectric materials, such as silicon oxide, silicon nitride, silicon oxynitride, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), low-k dielectric material, and/or other applicable dielectric materials. Examples of the low-k dielectric materials include, but are not limited to, fluorinated silica glass (FSG), carbon doped silicon oxide, amorphous fluorinated carbon, parylene, bis-benzocyclobutenes (BCB), or polyimide. In addition, the ILD layer **128** may be formed by chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), spin-on coating, or another applicable process.

Afterwards, a planarizing process is performed on the ILD layer **128** until the top surfaces of the dummy gate structures **120** are exposed, as shown in FIG. 1H in accordance with some embodiments. After the planarizing process, the top surfaces of the dummy gate structures **120** may be substantially level with the top surfaces of the gate spacers **122** and the ILD layer **128**. In some embodiments, the planarizing process includes a grinding process, a chemical mechanical polishing (CMP) process, an etching process, another applicable process, or a combination thereof.

Next, the dummy gate structures **120** are removed to form trenches **130** in the ILD layer **128**, as shown in FIG. 1I in accordance with some embodiments. More specifically, each of the trenches **130** is formed between each pair of the gate spacers **122**, and the fin structure **110** is exposed by the trenches **130**. The dummy gate dielectric layer **116** and the dummy gate electrode layer **118** are removed by an etching process, such as a dry etching process or a wet etching process.

After the trenches **130** are formed, gate dielectric layers **132** and gate electrode layers **134** (which may be collectively referred to as a gate structure) are formed in the trenches **130**, as shown in FIG. 1J in accordance with some embodiments. More specifically, the gate electrode layers **134** are formed over the gate dielectric layers **132**, and sidewalls of the gate electrode layers **132** may be covered by

the gate dielectric layers **132**. In addition, the gate structure may also include work function layers (not shown) formed between each of the gate dielectric layers **132** and each of the gate electrode layers **134**.

Each of the gate dielectric layers **132** may be a single layer or multiple layers. In some embodiments, the gate dielectric layers **132** are made of silicon oxide, silicon nitride, silicon oxynitride (SiON), dielectric material(s) with high dielectric constant (high-k), or a combination thereof. In some embodiments, the gate dielectric layers **132** are deposited using a plasma enhanced chemical vapor deposition (PECVD) process or a spin coating process.

Moreover, the gate electrode layers **134** are made of a conductive material such as aluminum (Al), copper (Cu), tungsten (W), titanium (Ti), tantalum (Ta), or another applicable material, in accordance with some embodiments. The gate electrode layers **134** may be formed by a deposition process, such as a chemical vapor deposition (CVD) process, a physical vapor deposition (PVD) process, an atomic layer deposition (ALD) process, a high density plasma CVD (HDPCVD) process, a metal organic CVD (MOCVD) process, or a plasma enhanced CVD (PECVD) process.

The work function layers may be made of metal materials, and the metal materials may include N-work-function metal or P-work-function metal. The N-work-function metal may include tungsten (W), copper (Cu), titanium (Ti), silver (Ag), aluminum (Al), titanium aluminum alloy (TiAl), titanium aluminum nitride (TiAlN), tantalum carbide (TaC), tantalum carbon nitride (TaCN), tantalum silicon nitride (TaSiN), manganese (Mn), zirconium (Zr) or a combination thereof. The P-work-function metal may include titanium nitride (TiN), tungsten nitride (WN), tantalum nitride (TaN), ruthenium (Ru) or a combination thereof.

FIGS. 2A-2I are cross-sectional views illustrating various stages of forming the semiconductor structure in accordance with some embodiments. It should be noted that FIGS. 2A-2I are illustrated along the line A-A' shown in FIG. 1J and the process shown in FIG. 2A continue from the process shown in the FIG. 1J. A mask layer **200** is formed on the ILD layer **128**, as shown in FIG. 2A in accordance with some embodiments. In some embodiments, the mask layer **200** is formed by deposition processes, which may include a chemical vapor deposition (CVD) process, a high-density plasma chemical vapor deposition (HDPCVD) process, a spin-on process, a sputtering process, or another applicable process. In some embodiments, the mask layer **200** is provided for an isolation process of the gate structure (such as the gate dielectric layers **132** and gate electrode layers **134** in FIG. 1J). To be more specific, a portion of the gate structure that is uncovered by the mask layer **200** may be etched and filled with isolation material. Otherwise, the covered gate structure is not removed.

Next, an etching process is performed to obtain a patterned mask layer **201** and a patterned ILD layer **129**, as shown in FIG. 2B in accordance with some embodiments. The etching process may be a dry etching process or a wet etching process. In some embodiments, the etching process is an anisotropic etching process. After the etching process is complete, a first opening **140** is formed in the patterned ILD layer **129** and over the second fin structure **110b** and the second epitaxial structure **124b**. In some embodiments, the first opening **140** penetrates the patterned ILD layer **129** and exposes the CESL **126**. In some embodiments, a bottom of the first opening **140** is directly above the second epitaxial structure **124b** and the top surface of one of the fin isolation structures **111**. That is, the bottom of the first opening **140**

overlaps with the second epitaxial structure **124b** and the top surface of the fin isolation structure **111** in a top view.

Afterwards, the patterned mask layer **201** is removed, as shown in FIG. 2C in accordance with some embodiments. In some embodiments, a dielectric material **142** is formed in the first opening **140** and around the patterned ILD layer **129**. For example, the material of the dielectric material **142** may include SiC, LaO, AlO, AON, ZrO, HfO, SiN, Si, ZnO, ZrN, ZrAlO, TiO, TaO, YO, TaCN, ZrSi, SiOCN, SiOC, SiCN, HfSi, or a combination thereof. In some embodiments, the dielectric material **142** is made of dielectric material(s) with high dielectric constant (high-k). In some embodiments, the dielectric material **142** is deposited using a plasma enhanced chemical vapor deposition (PECVD) process, a spin coating process, any other suitable method, or a combination thereof. Optionally, a planarization process may be performed so that the top surface of the dielectric material **142** may be level with the top surface of the patterned ILD layer **129**.

Next, a photoresist layer **300** is formed on the patterned ILD layer **129**, and the photoresist layer **300** exposes the dielectric material **142** in the first opening **140**, as shown in FIG. 2D in accordance with some embodiments. The photoresist layer **300** may be formed by a deposition process and a patterning process. The deposition process for forming the photoresist layer **300** may include a chemical vapor deposition (CVD) process, a high-density plasma chemical vapor deposition (HDPCVD) process, a spin-on process, a sputtering process, or another applicable process. The patterning process for forming the photoresist layer **300** may include a photolithography process and an etching process. The photolithography process may include photoresist coating (e.g., spin-on coating), soft baking, mask aligning, exposure, post-exposure baking, developing the photoresist, rinsing, and drying (e.g., hard baking). The etching process may include a dry etching process or a wet etching process.

After the photoresist layer **300** is formed, the exposed dielectric material **142** in the first opening **140** is partially removed using an etching process, as shown in FIG. 2E in accordance with some embodiments. As a result, a recessed dielectric material **143** is obtained, and the photoresist layer **300** is removed. The etching process may include a dry etching process or a wet etching process. In some embodiments, the top surface of the recessed dielectric material **143** is lower than the top surface of the patterned ILD layer **129**. In some embodiments, the top surface of the recessed dielectric material **143** is lower than the top surface of the dielectric material **142** outside the first opening **140** (i.e. covered by the photoresist layer **300**).

Then, the patterned ILD layer **129** that is adjacent to the recessed dielectric material **143** is removed by an etching process, as shown in FIG. 2F in accordance with some embodiments. Since the material of the recessed dielectric material **143** (and the dielectric material **142**) is different from the material of the patterned ILD layer **129**, the etch rate of the recessed dielectric material **143** is also different from the etch rate of the patterned ILD layer **129**. In some embodiments, the etch rate of the recessed dielectric material **143** is substantially slower than the etch rate of the patterned ILD layer **129**. Accordingly, all or most of the patterned ILD layer **129** is removed, while the recessed dielectric material **143** nearly remains its profile before the etching process performed.

In some embodiments, the CESL **126** serves as an etch stop layer of this etching process, and a portion of the CESL **126** located directly below the patterned ILD layer **129** is also removed, forming a second opening **150** for a contact

166 that is subsequently formed. The top surfaces of the first epitaxial structure 124a and the third epitaxial structure 124c are exposed in the second opening 150, and the second epitaxial structure 124b is covered and insulated by the recessed dielectric material 143. In some embodiments, the etching process may include a dry etching process or a wet etching process.

Next, a liner 162 is formed in the second opening 150 along the sidewalls of the dielectric material 142 and the sidewalls of the recessed dielectric material 143, as shown in FIG. 2G in accordance with some embodiments. In some embodiments, the material of the liner 162 includes SiC, LaO, AlO, AON, ZrO, HfO, SiN, Si, ZnO, ZrN, ZrAlO, TiO, TaO, YO, TaCN, ZrSi, SiOCN, SiOC, SiCN, HfSi, or a combination thereof. In some embodiments, the liner 162 extends onto the top surface of the first epitaxial structure 124a along a sidewall of one of the fin isolation structures 111. In some embodiments, the liner 162 ends at the top surface of one of the fin isolation structures 111 before reaching the top surface of the first epitaxial structure 124a.

It should be appreciated that the liner 162 may also extend onto the top surface of the third epitaxial structure 124c along a sidewall of one of the fin isolation structures 111, or end at the top surface of one of the fin isolation structures 111 before reaching the top surface of the third epitaxial structure 124c. Any possible configuration of the liner 162 is within the scope of the present disclosure. In some embodiments, the liner 162 is conformally formed along the sidewalls of the dielectric material 142 and the sidewalls of the recessed dielectric material 143. The width W2 of the liner 162 is about 1 nm to about 20 nm. Accordingly, it would leave enough space in the second opening 150 to form the contact 166 without increasing the difficulty of forming the liner 162.

Then, a silicide layer 164 is formed on the first epitaxial structure 124a and the third epitaxial structure 124c in the second opening 150 using a silicidation process, as shown in FIG. 2H in accordance with some embodiments. For example, the silicide layer 164 may be formed in a portion of the first epitaxial structure 124a and the third epitaxial structure 124c. In some embodiments, the silicidation process includes a metal material deposition process and an annealing process performed in sequence. In some embodiments, the deposition process of the silicidation process includes a physical vapor deposition (PVD) process, an atomic layer deposition (ALD) process, or another applicable process. In some embodiments, the annealing process of the silicidation process is performed at a temperature of about 300° C. to about 800° C. After the annealing process, the unreacted metal material is removed. In some embodiments, the silicide layer 164 is conformally formed and therefore a thickness of the silicide layer 164 is substantially uniform across the silicide layer 164. In some embodiments, the top surface of the silicide layer 164 is lower than the top surface of the fin isolation structures 111.

In some embodiments, the silicide layer 164 is formed of one or more of cobalt silicide (e.g. CoSi, CoSi₂, Co₂Si, Co₃Si; collectively “Co silicide”), titanium silicide (e.g. Ti₅Si₃, TiSi, TiSi₂, TiSi₃, Ti₆Si₄; collectively “Ti silicide”), nickel silicide (e.g. Ni₃Si, Ni₃₁Si₁₂, Ni₂Si, Ni₃Si₂, NiSi, NiSi₂; collectively “Ni silicide”), copper silicide (e.g. Cu₁₇Si₃, Cu₅₆Si₁₁, Cu₅Si, Cu₃₃Si₇, Cu₄Si, Cu₁₉Si₆, Cu₃Si, Cu₈₇Si₁₃; collectively “Cu silicide”), tungsten silicide (W₅Si₃, WSi₂; collectively “W silicide”), and molybdenum silicide (Mo₃Si, Mo₅Si₃, MoSi₂; collectively “Mo silicide”).

Afterwards, a contact 166 is filled into the second opening 150 and passes through the dielectric material 142, as shown

in FIG. 2I in accordance with some embodiments. In some embodiments, the contact 166 is formed over and electrically connected to the first epitaxial structure 124a and the third epitaxial structure 124c. In some embodiments, the top surface of the contact 166 may be leveled with the top surface of the dielectric material 142. For example, the contact 166 may be surrounded by the dielectric material 142. Moreover, the contact 166 may be electrically connected to the first epitaxial structure 124a and the third epitaxial structure 124c through the silicide layer 164. Since the recessed dielectric material 143 is formed on the second epitaxial structure 124b, the contact 166 is electrically insulated from the second epitaxial structure 124b, and therefore the second epitaxial structure 124b may be referred to as “the dummy epitaxial structure.”

In some embodiments, the thickness T of the recessed dielectric material 143 is about 2 nm to about 50 nm, wherein the thickness T may be measured from the top surface of the recessed dielectric material 143 to the top surface of the fin isolation structures 111. In some embodiments, the width W3 of the recessed dielectric material 143 is about 10 nm to about 100 nm, wherein the width W3 may be measured between the opposite sidewalls of the recessed dielectric material 143 between the first epitaxial structure 124a and the third epitaxial structure 124c. Accordingly, a stable insulation between the second epitaxial structure 124b and the contact 166 may be provided by the recessed dielectric material 143. Since the recessed dielectric material 143 is sandwiched between the second epitaxial structure 124b and the contact 166, the recessed dielectric material 143 may also be referred to as “the sandwiched portion” in the present disclosure.

In some embodiments, the contact 166 is in contact with the fin isolation structure 111. For example, the sidewall of the contact 166 intersects with the top surface of the fin isolation structure 111. An angle θ is formed between the top surface of the fin isolation structure 111 and the sidewall of the contact 166 via the recessed dielectric material 143. For example, the angle θ is an obtuse angle of about 90.5 degrees to about 110 degrees. As a result, the inclination between the top surface of the fin isolation structure 111 and the sidewall of the contact 166 would not increase the difficulty of the formation of the contact 166. In this way, the liner 162 and the contact 166 would fill the space between the fin isolation structures 111 over the first epitaxial structure 124a and/or the third epitaxial structure 124c, and the space is free from any dielectric material 142.

In some embodiments, the contact 166 is formed by deposition processes and a subsequent planarization process such as chemical mechanical polish (CMP). The contact 166 may be made of a conductive material, such as tungsten (W), cobalt (Co), ruthenium (Ru), copper (Cu), molybdenum (Mo), any other applicable material, or a combination thereof. In some embodiments, the contact 166 may be formed by any suitable deposition method, such as PVD, CVD, ALD, plating (e.g. electroplating). It should be appreciated that although one epitaxial structure (such as the second epitaxial structure 124b) is located below and insulated from the contact 166 in the present embodiment, it merely serves as an illustrative example. The number of the insulated epitaxial structure is not limited in the present application.

FIG. 3 is a cross-sectional view illustrating the semiconductor structure in accordance with some embodiments. It should be noted that the fin isolation structures 111 are omitted, as shown in FIG. 3 in accordance with some embodiments. The dielectric material 142 and the recessed

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dielectric material 143 are directly formed on the isolation structure 114, and surround the first epitaxial structure 124a, the second epitaxial structure 124b, and the third epitaxial structure 124c. In some embodiments, the liner 162 may extend onto the isolation structure 114 and the sidewall of the contact 166 ends at the top surface of the third epitaxial structure 124c. In some embodiments, the liner 162 and the contact 166 may extend onto the isolation structure 114 and located between the sidewalls of the first epitaxial structure 124a and the second epitaxial structure 124b. Since the fin isolation structures 111 are omitted, the number of steps for forming the semiconductor structure may be reduced and the overall process may be simplified.

FIG. 4 is a layout view illustrating the semiconductor structure in accordance with some embodiments. It should be noted that a conductive layer 180 and a conductive via 182 are formed over the contact 166 and configured to transmit signal from/to the first epitaxial structure 124a and the third epitaxial structure 124c, as shown in FIG. 4 in accordance with some embodiments. For example, the conductive via 182 is formed in a dielectric layer between the conductive layer 180 and the contact 166. In some embodiments, the contact 166 is electrically connected to the first epitaxial structure 124a and the third epitaxial structure 124c, simultaneously. As a result, the single conductive layer 180 may be electrically connected to the two epitaxial structures 124a and 124c via the conductive via 182 and the contact 166. Therefore, the overall conductive structure may be simplified, reducing the cost and complexity of the manufacturing process.

As described above, the present disclosure is directed to semiconductor structures and methods for forming the same are provided. The semiconductor structure may include a contact formed in the single layer and connected to the non-adjacent epitaxial structures. Accordingly, it is more likely to achieve miniaturization (for example, reducing cell height), and increase the routing flexibility. Moreover, the high-k dielectric material is adopted to secure the isolation between the contact and the dummy epitaxial structure.

In accordance with some embodiments, a semiconductor structure includes: a first fin structure, a second fin structure, and a third fin structure over a substrate, wherein the second fin structure is located between the first fin structure and the third fin structure. The semiconductor structure also includes a fin isolation structure formed between the first fin structure and the third fin structure; and a gate structure formed over the first fin structure, the second fin structure, the third fin structure and the fin isolation structure. The semiconductor structure further includes a first epitaxial structure formed over the first fin structure; a second epitaxial structure formed over the second fin structure; and a third epitaxial structure formed over the third fin structure. In addition the semiconductor structure includes: a dielectric material over the first epitaxial structure, the second epitaxial structure, and the third epitaxial structure; and a contact formed in the dielectric material and connected to the first epitaxial structure and the third epitaxial structure.

In accordance with some embodiments, a semiconductor structure includes: a plurality of fin structures over a substrate; and a gate structure formed over the fin structures. The semiconductor structure further includes a plurality of epitaxial structures formed over the fin structures. The semiconductor structure also includes a dielectric material over the epitaxial structures; and a contact formed in the dielectric layer, wherein the contact is located directly over and insulated from one of the epitaxial structures.

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In accordance with some embodiments, a method of forming a semiconductor structure includes: forming a first fin structure, a second fin structure and a third fin structure over a substrate, wherein the second fin structure is located between the first fin structure and the third fin structure. The method includes forming a fin isolation structure among the first fin structure, the second fin structure, and the third fin structure. The method also includes forming a gate structure over the first fin structure, the second fin structure, the third fin structure, and the isolation structure. The methods includes forming a first epitaxial structure over the first fin structure; forming a second epitaxial structure over the second fin structure; and forming a third epitaxial structure over the third fin structure. In addition, the method includes forming an interlayer dielectric layer over the first epitaxial structure, the second epitaxial structure, and the third epitaxial structure; and forming a first opening by etching the interlayer dielectric layer. The method includes filling a dielectric material into the first opening; and forming a second opening by etching the dielectric material and the interlayer dielectric layer, wherein the second opening exposes the first epitaxial structure and the third epitaxial structure. The method includes filling a conductive material into the second opening to form a contact, wherein the contact is connected to the first epitaxial structure and the third epitaxial structure.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A semiconductor structure, comprising:

- a first fin structure, a second fin structure, and a third fin structure over a substrate, wherein the second fin structure is located between the first fin structure and the third fin structure;
- a fin isolation structure formed between the first fin structure and the third fin structure;
- a gate structure formed over the first fin structure, the second fin structure, the third fin structure and the fin isolation structure;
- a first epitaxial structure formed over the first fin structure;
- a second epitaxial structure formed over the second fin structure;
- a third epitaxial structure formed over the third fin structure;
- a dielectric material over the first epitaxial structure, the second epitaxial structure, and the third epitaxial structure; and
- a contact formed in the dielectric material and connected to the first epitaxial structure and the third epitaxial structure, wherein the contact extends over the second epitaxial structures, a portion of the dielectric material is sandwiched between the second epitaxial structure and the contact, and a height of the sandwiched portion of the dielectric material is about 2 nm to about 50 nm.

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2. The semiconductor structure as claimed in claim 1, wherein the contact is in contact with the fin isolation structure.

3. The semiconductor structure as claimed in claim 2, wherein a sidewall of the contact intersects with a top surface of the fin isolation structure.

4. The semiconductor structure as claimed in claim 3, wherein an obtuse angle is formed between the top surface of the fin isolation structure and the sidewall of the contact via the dielectric material.

5. The semiconductor structure as claimed in claim 4, wherein the obtuse angle is less than or equal to 110 degrees.

6. The semiconductor structure as claimed in claim 3, wherein the top surface of the fin isolation structure is higher than top surfaces of the epitaxial structures.

7. The semiconductor structure as claimed in claim 1, wherein a width of the sandwiched portion of the dielectric material is about 10 nm to about 100 nm.

8. The semiconductor structure as claimed in claim 1, further comprising an interlayer dielectric layer formed over the first epitaxial structure, the second epitaxial structure, and the third epitaxial structure, and a dielectric constant of the dielectric material is different from a dielectric constant of the interlayer dielectric layer.

9. A semiconductor structure, comprising:
a plurality of fin structures over a substrate;
a gate structure formed over the fin structures;
a plurality of epitaxial structures formed over the fin structures;

a dielectric material over the epitaxial structures; and
a contact formed in the dielectric layer, wherein the contact is located directly over and insulated from one of the epitaxial structures, a portion of the dielectric material is sandwiched between the one of epitaxial structure and the contact, and a width of the sandwiched portion of the dielectric material is about 10 nm to about 100 nm.

10. The semiconductor structure as claimed in claim 9, wherein the contact is connected to two of the epitaxial structures.

11. The semiconductor structure as claimed in claim 10, wherein the two epitaxial structures are different types.

12. The semiconductor structure as claimed in claim 10, further comprising a silicide layer formed between the contact and the two epitaxial structures.

13. The semiconductor structure as claimed in claim 12, further comprising a fin isolation structure formed between the fin structures, wherein a top surface of the silicide layer is lower than the top surface of the fin isolation structure.

14. The semiconductor structure as claimed in claim 9, wherein a height of the sandwiched portion of the dielectric material is about 2 nm to about 50 nm.

15. A method of forming a semiconductor structure, comprising:

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forming a first fin structure, a second fin structure and a third fin structure over a substrate, wherein the second fin structure is located between the first fin structure and the third fin structure;

forming a fin isolation structure among the first fin structure, the second fin structure, and the third fin structure;

forming a gate structure over the first fin structure, the second fin structure, the third fin structure, and the isolation structure;

forming a first epitaxial structure over the first fin structure;

forming a second epitaxial structure over the second fin structure;

forming a third epitaxial structure over the third fin structure;

forming an interlayer dielectric layer over the first epitaxial structure, the second epitaxial structure, and the third epitaxial structure;

forming a first opening by etching the interlayer dielectric layer;

filling a dielectric material into the first opening;

forming a second opening by etching the dielectric material and the interlayer dielectric layer, wherein the second opening exposes the first epitaxial structure and the third epitaxial structure; and

filling a conductive material into the second opening to form a contact, wherein the contact is connected to the first epitaxial structure and the third epitaxial structure and in contact with the fin isolation structure, wherein a sidewall of the contact intersects with a top surface of the fin isolation structure, and the top surface of the fin isolation structure is higher than top surfaces of the epitaxial structures.

16. The method as claimed in claim 15, wherein forming the second opening comprises etching the dielectric material and the interlayer dielectric layer using different etching processes.

17. The method as claimed in claim 15, wherein the first opening overlaps the second epitaxial structure, and the second epitaxial structure is spaced apart from the second opening.

18. The method as claimed in claim 15, further comprising forming a liner along a sidewall of the dielectric material.

19. The method as claimed in claim 18, further comprising:

forming a silicide layer over the first epitaxial structure and the third epitaxial structure after forming the liner.

20. The method as claimed in claim 15, wherein a height of the sandwiched portion of the dielectric material is about 2 nm to about 50 nm.

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